Unit 2 · Session 2 · Thu, Oct 29, 2026

Musculoskeletal Pharmacology — Session 2

4 parts · every fact cited to its slide or textbook page. About 185 minutes of focused study.

Study plan

25 min
Read the guide
One pass, top to bottom. Bold = what the lecture emphasized.

75 min
Diagrams & tables
Redraw each flow from memory; cover a table column and recall it.

25 min
Cram sheet
Hide answers and quiz yourself. Repeat until it's automatic.

55 min
Self-test
Answer before you look. Read every explanation, even when right.

5 min
Objectives check
Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. Gout runs **purine → uric acid (xanthine oxidase) → urate crystals → neutrophil flare**; treat the **flare** with colchicine, an NSAID, or a corticosteroid, and **never start urate-lowering therapy during a flare**. S5; S9; S16; SG U2-R10

2. **Allopurinol/febuxostat** cut uric acid production (xanthine oxidase), **probenecid** raises excretion (URAT1), and **pegloticase** breaks uric acid down — watch HLA-B*58:01, febuxostat's CV boxed warning, and G6PD deficiency. S12; S13; S14; S15; SG U2-R9

3. Muscle relaxants sort by site: **central** (baclofen GABA-B, tizanidine α2, cyclobenzaprine brainstem) vs. **peripheral** (**dantrolene** on RyR1 in muscle; botulinum toxin at the junction). S23; S24; SG U2-R11

4. **Dantrolene** is the emergency drug for **malignant hyperthermia**; never stop intrathecal **baclofen** abruptly; check **LFTs** and CYP1A2 inhibitors with tizanidine; cyclobenzaprine is anticholinergic (Beers). S28; S30; S32; S34; S35

5. Bone is a **balance**: PTH answers low calcium via bone, kidney, and gut; **bisphosphonates** (first-line), **denosumab**, raloxifene, and calcitonin slow resorption, while **teriparatide/abaloparatide** and **romosozumab** build bone. S40; S43; S47; S54; S56; SG U2-R12

6. Oral bisphosphonates need a **full glass of water, fasting, and 30 minutes upright**; denosumab must not just be stopped, teriparatide is 2 years, and romosozumab carries an MI/stroke boxed warning. S52; S53; S55; SG U2-R13

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO2.1 Identify major pharmacologic classes used in the management of musculoskeletal conditions and describe their mechanisms of action.

Part 1 · Gout: Acute Flares & Urate-Lowering Therapy
Part 2 · Skeletal Muscle Relaxants
Part 3 · Bone Pharmacology & Osteoporosis

IO2.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat musculoskeletal conditions.

Part 1 · Gout: Acute Flares & Urate-Lowering Therapy
Part 2 · Skeletal Muscle Relaxants
Part 3 · Bone Pharmacology & Osteoporosis

IO2.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.

Part 1 · Gout: Acute Flares & Urate-Lowering Therapy
Part 2 · Skeletal Muscle Relaxants
Part 3 · Bone Pharmacology & Osteoporosis
Part 4 · Putting It All Together: Musculoskeletal Toolkit

IO2.4 Select appropriate pharmacologic therapy for common musculoskeletal conditions based on clinical presentation and patient-specific factors.

Part 1 · Gout: Acute Flares & Urate-Lowering Therapy
Part 3 · Bone Pharmacology & Osteoporosis
Part 4 · Putting It All Together: Musculoskeletal Toolkit

IO2.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in musculoskeletal conditions.

Part 1 · Gout: Acute Flares & Urate-Lowering Therapy
Part 2 · Skeletal Muscle Relaxants
Part 3 · Bone Pharmacology & Osteoporosis

IO2.6 Explain important medication-use and safety information to patients receiving pharmacotherapy for musculoskeletal.

Part 2 · Skeletal Muscle Relaxants
Part 3 · Bone Pharmacology & Osteoporosis

SO1 Explain the pharmacologic treatment of acute and chronic gout, including representative generic agents.

Part 1 · Gout: Acute Flares & Urate-Lowering Therapy

SO2 Classify skeletal muscle relaxants by mechanism and clinical use, including representative generic agents.

Part 2 · Skeletal Muscle Relaxants

SO3 Explain the pharmacologic management of osteoporosis, including representative generic agents.

Part 3 · Bone Pharmacology & Osteoporosis

SO4 Identify key contraindications, adverse effects, and drug interactions throughout.

Part 1 · Gout: Acute Flares & Urate-Lowering Therapy
Part 2 · Skeletal Muscle Relaxants
Part 3 · Bone Pharmacology & Osteoporosis

SO5 Apply the full musculoskeletal pharmacology toolkit to integrated clinical scenarios.

Part 4 · Putting It All Together: Musculoskeletal Toolkit

ABOUT THIS LECTURE

- PA Pharmacology Course · Lecture 2 · Session 2 of 2 — Musculoskeletal Pharmacology (Thursday, October 29): gout, skeletal muscle relaxants, and bone pharmacology. S1
- Recap: Session 1 (Oct 27) covered musculoskeletal pain pathways, acetaminophen, NSAIDs, and osteoarthritis/rheumatoid arthritis pharmacotherapy including DMARDs. S2; S2 (notes)
- Today picks up with gout (Section 5) and runs through skeletal muscle relaxants, bone pharmacology, integration, and the cumulative Lecture Recall Quiz. S2 (notes)
- Required reading: Katzung 16e — Ch. 27 (skeletal muscle relaxants), the gout part of Ch. 36, and Ch. 42 (bone mineral homeostasis). Syllabus
- **Original figures in this deck:** Figures 4–8 (Gout: From Purine Metabolism to Joint Inflammation; Skeletal Muscle Relaxants: Central vs. Peripheral Site of Action; Calcium Homeostasis: The PTH Feedback Loop; Bisphosphonate Dosing Frequency; How Osteoporosis Drugs Shift the Remodeling Balance) were generated by the course director using the AI tool Claude to illustrate standard, non-copyrighted pharmacologic concepts. They are original diagrams, not reproduced from Katzung's Basic and Clinical Pharmacology or any other copyrighted source. **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai S90
- **Drug profiles (study reference, S72–S89):** formulation, standard adult dose, indications, adverse effects, contraindications and toxicity/overdose for each drug — standard adult dosing for teaching; verify against current prescribing information. S72; S72 (notes); S73
- Abbreviation key: **PO = by mouth**; **IV = intravenous**; **SC = subcutaneous**; **BID/TID = twice / three times daily**; **ER = extended release**; **CNS = central nervous system**; **GABA = gamma-aminobutyric acid**; **CYP = cytochrome P450**; **P-gp = P-glycoprotein**; **HLA = human leukocyte antigen**; **URAT1 = urate transporter 1**; **G6PD = glucose-6-phosphate dehydrogenase**; **RyR1 = ryanodine receptor 1**; **RANKL = receptor activator of nuclear factor kappa-B ligand**; **PTH = parathyroid hormone**; **SERM = selective estrogen receptor modulator**; **VTE = venous thromboembolism**; **SSRI = selective serotonin reuptake inhibitor**; **MAOI = monoamine oxidase inhibitor**; **CrCl = creatinine clearance**; **OTC = over the counter**. S72; S72 (notes)

SLIDE DISCREPANCIES

- **Febuxostat CV risk:** the slides teach a boxed warning for cardiovascular death; the textbook says early MACE concerns were not borne out. Answer course exams with the slide version (boxed warning). S13; p. 697
- **Colchicine window:** slides say best within 24 hours of onset; the textbook cites a 36-hour guideline window. Both agree: start early. S11; p. 696

- **ULT prophylaxis:** slides give colchicine or an NSAID for 3–6 months; the textbook says 6 months or longer. Answer with the slides. S13; p. 695
- **Bisphosphonate absorption and holidays:** slides say under 1% absorbed and a holiday after 3–5 years of oral therapy; the textbook says under 10% absorbed and reassessment after 5 years (3 for zoledronate). S48; S53; p. 819; p. 820
- **Starting vs. stopping in a flare:** don’t *start* urate-lowering therapy during a flare (slides), but don’t *stop* allopurinol a patient already takes (textbook). S9; p. 696

1 PART 1 · DRUG CLASS

Gout: Acute Flares & Urate-Lowering Therapy

SO1 SO4 IO2.1 IO2.2 IO2.3 IO2.4 IO2.5

Class

- Gout is a crystal-driven inflammatory arthritis with **two very different treatment phases**. S3
- **The chain:** purine metabolism → uric acid → monosodium urate crystals → joint inflammation. Everything downstream — colchicine, allopurinol, febuxostat, probenecid — targets a specific point along this chain. S4; S4 (notes)
- **Purine breakdown:** hypoxanthine → xanthine → uric acid (via **xanthine oxidase**) → hyperuricemia → urate crystals in the joint. S5
- Crystals activate neutrophils and macrophages (**NLRP3, IL-1β**) → the acute flare, the end of the cascade. S5
- **Raise urate:** diets rich in purines and fructose; alcohol, especially beer (more production, less renal excretion). S5
- **Lower urate excretion:** diuretics, especially thiazides and loop agents; chronic kidney disease sustains hyperuricemia. S5
- Walk the full chain and sort the drugs into **two buckets:** “lowers uric acid” (allopurinol, febuxostat, probenecid) vs. “treats the acute flare” (colchicine, NSAIDs, corticosteroids) — the same acute-vs-chronic distinction made explicit next. S6; S6 (notes)
- **Acute flare** — goal: control pain and inflammation now. Agents: colchicine, NSAIDs, corticosteroids. S9
- **Chronic management** — goal: lower serum uric acid to prevent future flares. Agents: allopurinol, febuxostat (↓ production); probenecid (↑ excretion). S9

Prototype + other drugs named

- Colchicine — acute treatment (plant-derived alkaloid, no stem). S10; S19
- Allopurinol — first-line urate-lowering therapy for most patients. S12
- Febuxostat — alternative xanthine oxidase inhibitor. S12
- Probenecid — uricosuric (second-line oral). Pegloticase, rasburicase — uricases (refractory). S14; S15

- **TB** Humans lack uricase, the enzyme other mammals use to turn uric acid into soluble allantoin; most people with hyperuricemia never have a gout attack. p. 695
- **TB** Treat gout, not a number: long-term drug treatment of asymptomatic hyperuricemia is unproven. Adherence is poor (18–26% in younger patients). p. 695

Mechanism of action

- **Mechanism: disrupts microtubule formation in neutrophils.** Colchicine binds tubulin → less neutrophil migration and NLRP3 activation → less inflammation at the crystal site. S10; S11; SG U2-M
- Colchicine works on the cell's cytoskeleton, so its effect builds in the early hours of a flare. S11 (notes)
- **Allopurinol** (via its metabolite oxypurinol) and **febuxostat** block **xanthine oxidase** → ↓ uric acid **production**. S12; S13; SG U2-R9
- Febuxostat is also a xanthine oxidase inhibitor, with a different chemical structure. S12
- **Probenecid — mechanism: inhibits renal tubular reabsorption of uric acid, increasing its excretion** (blocks URAT1). A uricosuric, not a xanthine oxidase inhibitor — a different strategy. S14; S15; SG U2-R9
- **Pegloticase, rasburicase:** recombinant **uricase** converts uric acid to **allantoin**, which is soluble and excreted. S14; S15
- Uricosurics block reabsorption; uricases remove uric acid enzymatically, which is why they suit refractory disease. S15 (notes)
- NSAIDs and corticosteroids reduce joint inflammation in a flare. S6
- **TB** Allopurinol is a purine analog; its metabolite oxypurinol (half-life 15 hours) also inhibits xanthine oxidase, so it is dosed once daily. p. 695
- **TB** Colchicine relieves gout pain and inflammation in 12–24 hours without changing urate metabolism or excretion. p. 696

Indications

- **Treat the acute flare first** (colchicine, NSAID, or corticosteroid); urate-lowering therapy is deferred until the flare resolves, then used for long-term prevention. S16; SG U2-R10
- Colchicine is most effective when started early in a flare — **best within 24 hours of onset**. S10; S11
- Allopurinol is first-line for most patients; febuxostat is the alternative when allopurinol is not tolerated or contraindicated. S12
- Probenecid is an older, second-line oral option. **Pegloticase — a recombinant uricase enzyme for severe, refractory gout not controlled by other agents.** S14; S14 (notes); S15
- **TB** Colchicine flare regimen: 1.2 mg, then 0.6 mg an hour later; prophylaxis 0.6 mg once or twice daily during urate-lowering therapy initiation. p. 696
- **TB** Corticosteroids are a good alternative when NSAIDs or colchicine are contraindicated, including CKD: prednisone 30–50 mg/d for 1–2 days, tapered over 7–10 days, or intra-articular. p. 698
- **TB** All NSAIDs except aspirin, salicylates, and tolmetin treat gout flares. p. 697
- **TB** Febuxostat appears safe in moderate CKD; it can be tried after allopurinol hypersensitivity, though a reaction may still occur. p. 697

- TB Glucocorticoids in gout: reduce neutrophil migration, prostaglandins, and IL-1β. p. 698

Adverse effects (boxed warnings first)

- **Febuxostat: boxed warning for cardiovascular death** — it carries its own cardiovascular safety considerations. S12; S13
- **Colchicine: narrow therapeutic window** — GI adverse effects (diarrhea) are common and dose-limiting. S10; S11
- **Allopurinol:** the **HLA-B*58:01** allele raises the risk of severe hypersensitivity / severe cutaneous reactions. S12; S13
- Pegloticase: infusion reactions are common; anti-drug antibodies can reduce response. S15
- TB Allopurinol hypersensitivity syndrome (SJS/TEN, fever, eosinophilia, hepatitis, renal failure) usually appears 2–4 weeks after starting — stop at the first rash. p. 696
- TB Colchicine can also cause marrow suppression, myopathy, rhabdomyolysis, and neuritis; it is not removed by dialysis. p. 696
- TB Febuxostat: liver function abnormalities, diarrhea, headache, nausea. p. 697

Contraindications

- **Do NOT start urate-lowering therapy during an acute flare** — starting or changing it then can actually prolong the flare. Start or adjust only once the flare has resolved. S9; S9 (notes); SG U2-R10
- **Probenecid:** avoid with a history of uric acid kidney stones or significant renal impairment; needs adequate hydration. S14; S15
- **Pegloticase and rasburicase:** contraindicated in **G6PD deficiency** (hemolysis risk). S15
- TB Aspirin is avoided in gout: at low doses (≤2.6 g/d) it causes renal retention of uric acid. p. 697
- TB Uricosurics should not be used in patients who already excrete large amounts of uric acid. p. 698

Drug interactions

- **Colchicine + CYP3A4 / P-gp inhibitors** (clarithromycin, cyclosporine, strong CYP3A4 inhibitors) → raised levels and toxicity. S11
- **Allopurinol + azathioprine or mercaptopurine:** reduce the dose (xanthine oxidase also clears these drugs). S13
- TB Allopurinol with azathioprine: cut the azathioprine dose by 75%. p. 696
- TB Probenecid also reduces renal secretion of other weak acids, such as penicillin. p. 698

Pharmacokinetics

- Colchicine: adjust the dose for renal or hepatic impairment. S11
- Allopurinol: reduce the dose in renal impairment. S13
- Pegloticase is given by IV infusion. S14
- TB Allopurinol: start 100 mg/d (≤50 mg/d in CKD), titrate every 2–5 weeks to uric acid <6 mg/dL (max 800 mg/d). p. 695

Monitoring

- Xanthine oxidase inhibitors: **start low and titrate upward toward a serum urate target**. S13
- When starting urate-lowering therapy, give **colchicine or an NSAID as prophylaxis for 3–6 months**. S13
- **TB** Probenecid: keep urine volume high and urine pH above 6.0 early on to prevent stones; start 2–3 weeks after an attack. p. 698
- **TB** Pegloticase: premedicate with a corticosteroid and antihistamine; flares are common in the first 3–6 months. p. 698

CLINICAL PEARL / WHY

- This acute-vs-chronic distinction is one of the single most testable concepts in gout pharmacology. S9 (notes)
- Allopurinol vs. febuxostat: both sit on the same enzyme; the choice turns on safety profile and tolerability. S13 (notes)
- **HLA-B*5801** is a direct callback to Lecture 1's pharmacogenomics section. S12 (notes)
- Pegloticase is reserved for refractory cases given its cost, infusion-reaction risk, and need for specialty administration. S14
- Case: sudden, severe pain, redness, and swelling of the first metatarsophalangeal joint — an acute gout flare. **Is this the moment to start acute treatment, chronic urate-lowering therapy, or both?** Acute treatment now; ULT later. S16
- **Name clues:** suffixes often reveal the mechanism before you recall the drug — use them as memory hooks, then confirm each class before prescribing. S19; S19 (notes)
- **-ostat** marks xanthine oxidase inhibitors (allopurinol is the exception); **-ase** names enzymes, so pegloticase and rasburicase are uricases; **-sone, -olone** mark corticosteroids used to treat flares. S19
- **TB** Allopurinol already running when a flare hits? Don't stop it — stopping can worsen and lengthen the attack. p. 696

Step by step

From purines to a flare S5

Purine breakdown → Hypoxanthine → xanthine →

Uric acid (xanthine oxidase) → Hyperuricemia →

Monosodium urate crystals in joint →

Neutrophil/macrophage activation (NLRP3, IL-1β) → Acute flare

Gout visit decision S9; S13; S15; S16

Acute flare? → Yes → colchicine, NSAID, or corticosteroid now →

Defer urate-lowering therapy →

Flare resolved → start allopurinol low, titrate to urate target →

Cover with colchicine/NSAID prophylaxis 3–6 months →

Refractory → pegloticase (check G6PD)

How colchicine works S11

Colchicine binds tubulin → Blocks microtubule polymerization →

Reduced neutrophil migration and NLRP3 activation →

Less inflammation at the crystal site

Figure 4 · Gout: from purine metabolism to joint inflammation. Original illustrative diagram (not from Katzung): purine metabolism → xanthine → uric acid → monosodium urate (MSU) crystals in the joint → neutrophil-driven joint inflammation. Allopurinol and febuxostat (xanthine oxidase inhibitors) reduce uric acid production; probenecid increases renal excretion of uric acid; colchicine, NSAIDs, and corticosteroids reduce joint inflammation. S6; S6 (notes)

Acute flare vs. chronic urate-lowering therapy

	Acute flare	Chronic management
Goal	Control pain and inflammation **now**	Lower serum uric acid to **prevent** future flares
Agents	Colchicine, NSAIDs, corticosteroids	Allopurinol, febuxostat (↓ production); probenecid (↑ excretion)
When	At once — colchicine best within 24 h	Only after the flare resolves; start low, titrate to a urate target
Trap	ULT is not a flare treatment	Starting/changing ULT in a flare can prolong it; cover with prophylaxis 3–6 months

How to tell them apart. Sources: S9; S11; S13; SG U2-R10

The gout drugs side by side

Drug	Mechanism	Key safety point
Colchicine	Disrupts neutrophil microtubules	Narrow window; diarrhea dose-limiting; CYP3A4/P-gp inhibitors; renal/hepatic dosing

Drug	Mechanism	Key safety point
Allopurinol	Xanthine oxidase inhibitor (via oxypurinol)	HLA-B*58:01 hypersensitivity; renal dosing; ↓ dose with azathioprine/mercaptopurine
Febuxostat	Xanthine oxidase inhibitor (different structure)	Boxed warning: cardiovascular death
Probenecid	Uricosuric: blocks URAT1 reabsorption	Hydrate; avoid with uric acid stones or significant renal impairment
Pegloticase, rasburicase	Uricase: uric acid → allantoin	G6PD deficiency (hemolysis); infusion reactions; antibodies

How to tell them apart. Sources: S10–S15

Name clues: gout

Name pattern	Drug class	Examples
-ostat	Xanthine oxidase inhibitor	Febuxostat (allopurinol is the exception)
-ase	Enzyme (uricase)	Pegloticase, rasburicase
colchicine	Plant-derived alkaloid, no stem	Colchicine
probenecid, lesinurad-type	Uricosurics	Probenecid
-profen	Propionic-acid NSAIDs	Ibuprofen, naproxen
-sone, -olone	Corticosteroids	Prednisone, methylprednisolone

How to tell them apart. Sources: S19

Gout drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Colchicine (Colcrys, Mitigare)	**Anti-inflammatory (microtubule inhibitor)**	Tablets 0.6 mg; capsules 0.6 mg	Flare: 1.2 mg PO then 0.6 mg one hour later, within 24 hours of onset; prophylaxis: 0.6 mg once or twice daily	Acute gout flare, flare prophylaxis, familial Mediterranean fever
Allopurinol (Zyloprim)	**Xanthine oxidase inhibitor**	Tablets 100/300 mg	Start 50-100 mg PO daily, titrate every 2-5 weeks to serum urate under 6 mg/dL (usual 300-800 mg/day)	Chronic gout, hyperuricemia, uric acid stones, tumor lysis prophylaxis
Febuxostat (Uloric)	**Xanthine oxidase inhibitor (non-purine)**	Tablets 40/80 mg	40 mg PO daily, increase to 80 mg if urate above 6 mg/dL after 2 weeks	Chronic gout when allopurinol fails or is not tolerated

Drug	Class	Formulation	Adult dose	Indications
Probenecid	**Uricosuric (URAT1 inhibitor)**	Tablets 500 mg	250 mg PO twice daily for 1 week, then 500 mg twice daily; maximum 2 g/day	Chronic gout with underexcretion of urate; adjunct to penicillin
Pegloticase (Krystexxa)	**Recombinant uricase**	IV 8 mg/mL vial	8 mg IV every 2 weeks over at least 2 hours, with methotrexate co-therapy	Refractory chronic gout

How to tell them apart. Sources: S73; S74; S75; S76; S77

Gout drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Colchicine (Colcrys, Mitigare)	Diarrhea, nausea, vomiting, abdominal pain; myelosuppression and myopathy with toxicity	Severe renal or hepatic impairment with CYP3A4 or P-gp inhibitors; blood dyscrasias	Narrow therapeutic window: severe GI toxicity, bone marrow suppression, multi-organ failure; avoid clarithromycin and strong CYP3A4/P-gp inhibitors
Allopurinol (Zyloprim)	Rash, GI upset, flare at initiation, elevated liver enzymes	Prior severe reaction; HLA-B*58:01 carriers (higher risk); reduce dose in renal impairment	Allopurinol hypersensitivity syndrome (fever, rash, eosinophilia, organ injury); interacts with azathioprine and mercaptopurine
Febuxostat (Uloric)	Gout flare at initiation, nausea, rash, elevated liver enzymes, cardiovascular death	Concurrent azathioprine or mercaptopurine; caution with cardiovascular disease	Boxed warning for cardiovascular death; monitor liver tests
Probenecid	GI upset, rash, headache, kidney stones, flare at start	Uric acid kidney stones, significant renal impairment (eGFR under 30), blood dyscrasias, during acute flare	Renal colic and stones; hydrate well and alkalinize urine; raises levels of penicillins, methotrexate and NSAIDs
Pegloticase (Krystexxa)	Infusion reaction, gout flares, anaphylaxis, nausea, anti-drug antibodies	G6PD deficiency; uncontrolled hypertension (caution); stop other urate-lowering drugs	Boxed warning for anaphylaxis, infusion reactions, hemolysis and methemoglobinemia in G6PD deficiency; premedicate and monitor serum urate

How to tell them apart. Sources: S73; S74; S75; S76; S77

Gout drugs in the reading (textbook)

Drug	Dose / PK	What sets it apart
Allopurinol	Start 100 mg/d; +100 mg every 2–5 weeks to <6 mg/dL (max 800); oxypurinol t½ 15 h	Pruritic rash 3%; AHS 2–4 weeks in; HLA-B 5801 (Han Chinese, Korean, Thai); cut azathioprine 75%
Colchicine	1.2 mg then 0.6 mg; prophylaxis 0.6 mg once–twice daily; t½ 9 h	Within 36 h (guideline); not dialyzable; no longer given IV
Febuxostat	40 or 80 mg daily; target <6.0 mg/dL	Safe in moderate CKD; for allopurinol intolerance
NSAIDs	Any except aspirin, salicylates, tolmetin	Aspirin ≤2.6 g/d retains urate
Pegloticase	8 mg IV every 2 weeks; t½ 6–14 days	With methotrexate fewer reactions; flares first 3–6 months; avoid in G6PD deficiency
Probenecid (lesinurad withdrawn 2019)	0.5 g/d → 1 g/d after 1 week	Start 2–3 weeks after an attack; high urine volume, pH > 6.0; ↓ penicillin secretion
Glucocorticoids in gout	Prednisone 30–50 mg/d × 1–2 days, taper 7–10 days; intra-articular triamcinolone	Rule out joint infection; good in CKD
Interleukin-1 inhibitors in gout	Anakinra 100 mg/day × 3 days; canakinumab 150 mg once	Not FDA approved for gout; option when standard drugs fail

How to tell them apart. Sources: p. 695; p. 696; p. 697; p. 698

SLIDE DISCREPANCY

- **Febuxostat cardiovascular risk**
- Slides: Febuxostat carries a boxed warning for cardiovascular death. S13
- Textbook: Initial concerns about increased MACE with febuxostat have not been borne out; the FDA approved only the 40- and 80-mg doses. p. 697

SLIDE DISCREPANCY

- **Colchicine timing**
- Slides: Best started within 24 hours of flare onset. S11
- Textbook: Guidelines suggest giving colchicine within 36 hours of an attack; efficacy falls after that. p. 696

WHY IT HAPPENS

- Why not start allopurinol in a flare? Shifting urate levels mobilizes crystals — starting or changing ULT during a flare can worsen or prolong it. Calm the flare first. S9 (notes); SG U2-R10

- Why does colchicine have to start early? It works on the neutrophil's cytoskeleton, stopping migration — once neutrophils have flooded the joint, it has less to prevent. S11; S11 (notes)

MNEMONIC (MEMORY AID)

- **Make less, pee more, break it down:** xanthine oxidase inhibitors make less; probenecid pees more; pegloticase breaks it down. S12–S15; SG U2-R9

HIGH-YIELD

- Acute gout treatment and chronic urate-lowering therapy are different goals, started at different times. S67; S9
- Flare: colchicine, NSAID, or corticosteroid. Prevention: allopurinol/febuxostat (↓ production) or probenecid (↑ excretion). S9; SG U2-R9
- Colchicine: microtubules; best within 24 h; diarrhea; CYP3A4/P-gp inhibitors. S10; S11
- Allopurinol: HLA-B*58:01; febuxostat: boxed CV death warning. S12; S13
- Pegloticase/rasburicase: no G6PD deficiency. S15

2 PART 2 · DRUG CLASS

Skeletal Muscle Relaxants

SO2 SO4 IO2.1 IO2.2 IO2.3 IO2.5 IO2.6

Class

- Different mechanisms for **spasm, spasticity, and rigidity** — not the same thing, and they often require distinct pharmacologic approaches. S20; S21
- **Muscle spasm:** sudden, involuntary contraction, often from acute musculoskeletal injury; typically treated with centrally acting relaxants. S21
- **Spasticity:** velocity-dependent increased tone from **upper motor neuron injury**. **Rigidity:** constant increased tone, independent of velocity. S21
- **Where tone is controlled:** upper motor neurons (descending inhibitory pathways) ▸ spinal interneurons (GABA, glycine, alpha-2) ▸ alpha motor neuron ▸ neuromuscular junction (ACh release) ▸ muscle fiber contraction (SR Ca^{2+} release). S22
- **Centrally acting** (act within the CNS to reduce muscle tone): cyclobenzaprine, tizanidine, baclofen — first-line; carisoprodol, methocarbamol, metaxalone, chlorzoxazone — other options. S23
- **Peripherally / locally acting:** dantrolene acts directly on skeletal muscle itself; botulinum toxin acts at the neuromuscular junction — useful when a central mechanism is not desired (or not safe). S23
- **Highest-yield distinction:** "where does the drug act?" sorts every agent into one of two buckets — **dantrolene is the only one in the peripheral (muscle) bucket.** S24; S24 (notes); SG U2-R11
- **TB** Spasticity is common after spinal injury, cerebral palsy, multiple sclerosis, and stroke;

Prototype + other drugs named

- Baclofen — GABA-B agonist (spinal cord). S27
- Tizanidine — central α2-adrenergic agonist. S29
- Cyclobenzaprine — tricyclic-like central relaxant (brainstem). S31
- Dantrolene — RyR1 blocker in the muscle fiber. S34

these drugs relieve painful spasm better than they restore function. p. 510; p. 511

Mechanism of action

- **Central agents** reduce CNS excitatory output to motor neurons → ↓ muscle tone. Baclofen and tizanidine act on spinal interneurons and afferents; cyclobenzaprine acts in the brainstem. S22; S24; SG U2-R11
- **Baclofen — mechanism: GABA-B receptor agonist in the spinal cord.** Binding GABA-B → opens K^+ channels, closes Ca^{2+} channels (presynaptic inhibition) → less excitatory transmitter release → lower motor neuron firing → reduced spasticity. S27; S28; SG U2-M
- **Tizanidine — mechanism: central α2-adrenergic agonist.** Activating alpha-2 receptors → reduced presynaptic release of excitatory amino acids → lower motor neuron activity → reduced muscle tone. S29; S30
- **Cyclobenzaprine:** acts in the brainstem; antagonizes **5-HT2** and descending serotonergic and noradrenergic pathways → reduced tonic somatic motor activity → muscle relaxation. Structurally related to tricyclic antidepressants. S31; S32
- **Dantrolene — mechanism: decreases calcium release from the skeletal muscle sarcoplasmic reticulum.** Nerve impulse → DHPR activated → **RyR1** opens → Ca^{2+} released → contraction; dantrolene blocks RyR1 → less Ca^{2+} → relaxation. S34; S35; S22; SG U2-R11; SG U2-M
- Dantrolene acts inside the fiber on RyR1 — directly on the muscle fiber, not the CNS — distinct from every centrally acting agent. S34; S22
- **Botulinum toxin — blocks presynaptic acetylcholine release at the neuromuscular junction** (ACh release at the junction) — a third mechanism, distinct from both CNS-acting drugs and dantrolene's sarcoplasmic-reticulum action. S33; S33 (notes); S22
- **TB** Diazepam (GABA-A) reduces spasticity partly in the spinal cord, but sedates at the doses needed. p. 511

Indications

- **Baclofen:** spasticity, particularly from spinal cord injury or multiple sclerosis; intrathecal option for severe spasticity. S27; S28
- Cyclobenzaprine — **primary use: short-term treatment of acute, localized muscle spasm**; short-term use only, 2–3 weeks. S31; S32
- Dantrolene — **clinical use #1: malignant hyperthermia (a true emergency indication)** (triggered by volatile anesthetics and succinylcholine) — a true emergency, perioperative indication where the drug is literally life-saving. S34; S34 (notes); S35; SG U2-R11
- **Dantrolene #2:** chronic spasticity, when central agents are not appropriate; also used for neuroleptic malignant syndrome. IV formulation for acute crisis. S34; S35
- **Botulinum toxin:** focal spasticity or dystonia; effect is local and temporary, requiring repeat injections. S33
- **TB** Malignant hyperthermia: give IV dantrolene 1 mg/kg, repeated as needed, while controlling acidosis and temperature. p. 513
- **TB** Cyclobenzaprine is ineffective for spasm from cerebral palsy or spinal cord injury. p. 513
- **TB** Gabapentin has shown promise as a spasmolytic in multiple sclerosis. p. 512

- TB Dantrolene binds RyR1; cardiac and smooth muscle use RyR2, so they are minimally depressed. p. 513

Adverse effects (boxed warnings first)

- TB Botulinum toxin: black box warning for distant spread of toxin, greatest in children treated for spasticity. p. 513
- **Baclofen:** CNS depression — sedation and weakness, additive risk with other CNS depressants. S27; S28
- **Baclofen withdrawal:** abrupt discontinuation (especially intrathecal) can cause a dangerous syndrome — **seizures, hyperthermia, rebound spasticity**. S27; S28
- **Tizanidine:** sedation, hypotension and bradycardia, dry mouth, **hepatotoxicity**. S29; S30
- **Cyclobenzaprine:** prominent sedation (commonly dosed at night); **anticholinergic** effects — dry mouth, blurred vision, urinary retention, confusion. S31; S32
- **Dantrolene:** hepatotoxicity; generalized weakness. S35
- **Carisoprodol — metabolized to meprobamate; Schedule IV controlled substance with dependence/misuse potential.** S33
- **Chlorzoxazone:** rare but notable hepatotoxicity risk. Methocarbamol is generally less sedating than cyclobenzaprine; metaxalone has a relatively favorable sedation profile. S33
- TB Tizanidine causes hypotension in 16–33%; abrupt withdrawal causes rebound hypertension and tachycardia. p. 512
- TB High-dose baclofen: somnolence, respiratory depression, coma; more seizures in epilepsy. p. 512
- TB Carisoprodol withdrawal after long use: anxiety, tremor, twitching, insomnia, hallucinations. p. 511

Contraindications

- Cyclobenzaprine: use caution in older adults (**Beers list**); MAOIs contraindicated. S31; S32
- Tizanidine with ciprofloxacin or fluvoxamine is contraindicated (CYP1A2). S30

Drug interactions

- **Tizanidine + CYP1A2 inhibitors** (ciprofloxacin, fluvoxamine) → strongly raised levels; additive effect with antihypertensives. S30
- **Cyclobenzaprine:** serotonin syndrome risk with SSRIs, SNRIs, tramadol; MAOIs contraindicated. S32
- **Baclofen:** additive sedation with other CNS depressants. S27

Pharmacokinetics

- Baclofen: renal dose adjustment. S28
- **TB** Tizanidine: start 2 mg every 6–8 hours, max 36 mg/d; adjust for hepatic or renal impairment. p. 512
- **TB** Dantrolene: only about one-third of an oral dose is absorbed; half-life about 8 hours. p. 513

Monitoring

- Tizanidine requires liver function monitoring (LFTs) with regular use. S29; S30
- Never stop intrathecal baclofen abruptly. S27; S28
- **TB** Botulinum toxin injections: no more often than every 3 months. p. 513

CLINICAL PEARL / WHY

- Name clues: **-curonium / -curium** nondepolarizing neuromuscular blockers (vecuronium, atracurium, cisatracurium); **succinylcholine** depolarizing blocker; **-azepam** benzodiazepine (GABA-A, diazepam). S38
- Name clues: **cyclo-** tricyclic-like (cyclobenzaprine); **-idine** central alpha-2 agonist (tizanidine); baclofen has no shared stem (GABA analog); **-trolene** blocks RyR1 (dantrolene); **-toxin** blocks ACh release (botulinum toxin). S38
- **TB** Neuromuscular blockers (anesthesia, ICU) paralyze at the end plate and have no CNS activity; succinylcholine is the only depolarizing one. p. 499; p. 503
- **TB** Succinylcholine can cause fatal hyperkalemia after burns, denervation, or major trauma, and with volatile anesthetics can trigger malignant hyperthermia. p. 508
- **TB** Sugammadex reverses rocuronium/vecuronium; it binds progestins, so use backup contraception for 7 days. p. 509; p. 510

Step by step

Where muscle tone is controlled S22

Upper motor neurons (descending inhibition) →

Spinal interneurons (GABA, glycine, α2) — baclofen, tizanidine →

Alpha motor neuron → Neuromuscular junction (ACh) — botulinum toxin →

Muscle fiber (SR Ca^{2+} release) — dantrolene

Baclofen S28

Binds GABA-B receptors →

Opens K^+, closes Ca^{2+} channels (presynaptic inhibition) →

Less excitatory transmitter → Lower motor neuron firing → less spasticity

Dantrolene S35

Action potential → DHPR activated → RyR1 opens → Ca^{2+} released from SR →

Contraction → Dantrolene blocks RyR1 → less Ca^{2+} → Relaxation

Figure 5 · Skeletal muscle relaxants: central vs. peripheral site of action. Original illustrative diagram (not from Katzung): cyclobenzaprine (brainstem), tizanidine (α2), and baclofen (GABA-B, spinal cord) act within the central nervous system to reduce excitatory output to motor neurons, lowering muscle tone (spasm/spasticity); dantrolene acts directly on the skeletal muscle fiber itself, reducing Ca^{2+} release from the sarcoplasmic reticulum and muscle contraction. S24; S24 (notes)

Spasm vs. spasticity vs. rigidity

	What it is	Typical approach
Muscle spasm	Sudden, involuntary contraction, often after acute injury	Centrally acting relaxant (e.g., cyclobenzaprine, short term)
Spasticity	Velocity-dependent ↑ tone from upper motor neuron injury	Baclofen, tizanidine; dantrolene if central agents unsuitable; botulinum toxin if focal
Rigidity	Constant ↑ tone, independent of velocity	Distinct pharmacologic approach

How to tell them apart. Sources: S21; S23; S27; S33; S34

The muscle relaxants side by side

Drug	Site · mechanism	Use	Watch for
Baclofen	Spinal cord · GABA-B agonist	Spasticity (MS, spinal cord injury); intrathecal for severe	Sedation, weakness; renal dosing; **abrupt withdrawal: seizures, hyperthermia**

Drug	Site · mechanism	Use	Watch for
Tizanidine	CNS · α2 agonist	Spasticity / tone	Hypotension, bradycardia, dry mouth, **LFTs**; CYP1A2 (ciprofloxacin, fluvoxamine)
Cyclobenzaprine	Brainstem · 5-HT2 antagonism (TCA-like)	Acute local spasm, 2–3 weeks	Sedation, **anticholinergic** (Beers); serotonin syndrome; no MAOIs
Carisoprodol	CNS (→ meprobamate)	Other central option	Schedule IV: dependence, misuse
Methocarbamol · metaxalone	CNS	Other central options	Less sedating than cyclobenzaprine
Chlorzoxazone	CNS	Other central option	Rare hepatotoxicity
Dantrolene	Muscle fiber · blocks RyR1 Ca^{2+} release	**Malignant hyperthermia**; spasticity; NMS	Hepatotoxicity, weakness
Botulinum toxin	Neuromuscular junction · blocks ACh release	Focal spasticity, dystonia	Local and temporary; repeat injections

How to tell them apart. Sources: S23; S27–S35

Name clues: muscle relaxants

Name clue	Class	Examples
-curonium / -curium	Nondepolarizing neuromuscular blockers	Vecuronium, atracurium, cisatracurium
succinylcholine (-choline)	Depolarizing neuromuscular blocker	Succinylcholine
-azepam	Benzodiazepine (GABA-A)	Diazepam
cyclo- (tricyclic ring)	Tricyclic-like central relaxant	Cyclobenzaprine
-idine (imidazoline)	Central alpha-2 agonist	Tizanidine
No shared stem	GABA-B agonist (GABA analog)	Baclofen
-trolene (hydantoin derivative)	Blocks RyR1 in the muscle fiber	Dantrolene
-toxin	Blocks presynaptic ACh release	Botulinum toxin

How to tell them apart. Sources: S38

Skeletal muscle relaxant drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Baclofen (Lioresal, Gablofen)	**Centrally acting relaxant (GABA-B agonist)**	Tablets 5/10/20 mg; oral solution; intrathecal pump	5 mg PO three times daily, increase every 3 days to 40-80 mg/day in divided doses	Spasticity from spinal cord injury or multiple sclerosis
Tizanidine (Zanaflex)	**Centrally acting relaxant (alpha-2 agonist)**	Tablets 2/4 mg; capsules 2/4/6 mg	2 mg PO every 6-8 hours, titrate to 4-8 mg every 6-8 hours (maximum 36 mg/day)	Spasticity, off-label for muscle spasm
Cyclobenzaprine (Flexeril)	**Centrally acting relaxant (tricyclic-related)**	Tablets 5/7.5/10 mg; ER capsules 15/30 mg	5-10 mg PO three times daily for 2-3 weeks (ER 15-30 mg daily)	Acute muscle spasm from musculoskeletal conditions
Carisoprodol (Soma)	**Centrally acting relaxant (meprobamate metabolite)**	Tablets 250/350 mg	250-350 mg PO three times daily and at bedtime for up to 2-3 weeks	Acute painful musculoskeletal conditions
Methocarbamol (Robaxin)	**Centrally acting relaxant**	Tablets 500/750 mg; injection	1,500 mg PO four times daily for 2-3 days, then 750 mg four times daily	Acute painful musculoskeletal conditions
Dantrolene (Dantrium, Ryanodex)	**Direct-acting relaxant (RyR1 blocker)**	Capsules 25/50/100 mg; IV powder	Spasticity: 25 mg PO daily, titrate to 100 mg up to four times daily; malignant hyperthermia: 2.5 mg/kg IV, repeat to 10 mg/kg	Malignant hyperthermia, chronic spasticity, neuroleptic malignant syndrome

How to tell them apart. Sources: S78; S79; S80; S81; S82; S83

Skeletal muscle relaxant drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Baclofen (Lioresal, Gablofen)	Sedation, dizziness, weakness, nausea, confusion	Caution with renal impairment; avoid abrupt discontinuation	Overdose: coma, respiratory depression; abrupt withdrawal causes seizures, hyperthermia and rebound spasticity
Tizanidine (Zanaflex)	Sedation, dry mouth, hypotension, bradycardia, hepatotoxicity	Strong CYP1A2 inhibitors (ciprofloxacin, fluvoxamine); caution with antihypertensives and liver disease	Hypotension, bradycardia and CNS depression; check liver tests

Drug	Adverse effects	Contraindications	Toxicity / overdose
Cyclobenzaprine (Flexeril)	Sedation, dry mouth, dizziness, blurred vision, urinary retention, confusion	MAOIs within 14 days, arrhythmias, recent myocardial infarction, hyperthyroidism; Beers list for older adults	Serotonin syndrome with SSRIs and tramadol; tricyclic-type toxicity with arrhythmia, seizures in overdose
Carisoprodol (Soma)	Drowsiness, dizziness, headache, dependence	Acute intermittent porphyria; prior hypersensitivity; Schedule IV controlled substance	CNS depression, especially with opioids and alcohol; dependence and withdrawal
Methocarbamol (Robaxin)	Drowsiness, dizziness, nausea, urine discoloration	Hypersensitivity; IV form with renal impairment (polyethylene glycol)	CNS depression; supportive care
Dantrolene (Dantrium, Ryanodex)	Weakness, drowsiness, hepatotoxicity, diarrhea	Active liver disease (oral use); caution with calcium channel blockers (IV)	Boxed warning for hepatotoxicity (oral); generalized weakness; monitor liver enzymes

How to tell them apart. Sources: S78; S79; S80; S81; S82; S83

Spasmolytics in the reading (textbook)

Drug	Dose / PK	What sets it apart
Baclofen	15 mg twice daily → 100 mg daily; t½ 3–4 h; intrathecal	Less sedating than diazepam; withdraw very slowly; caution in pregnancy
Tizanidine	2 mg every 6–8 h, max 36 mg/d	1/10–1/15 of clonidine's BP effect; hypotension 16–33%; CYP1A2; rebound hypertension
Diazepam	4 mg/d → max 60 mg/d	GABA-A; works for spasm of any origin but sedates
Cyclobenzaprine	20–40 mg/d in divided doses	Antimuscarinic; confusion, hallucinations; not for CP or cord injury
Carisoprodol, meprobamate	Schedule IV	Withdrawal syndrome after extensive use
Botulinum toxin	Injected; lasts weeks to months	Cleaves SNAP-25/synaptobrevin; dystonia, spastic disorders; black box: spread of toxin
Dantrolene	25 mg/d → 100 mg four times daily; MH 1 mg/kg IV	RyR1 only; weakness, sedation, hepatitis
Gabapentin, pregabalin	Not covered in materials	Other centrally acting spasmolytic drugs: promising in MS spasticity

How to tell them apart. Sources: p. 511; p. 512; p. 513

Neuromuscular blockers (anesthesia) — the reading's essentials

	Nondepolarizing (rocuronium, vecuronium, cisatracurium)	Depolarizing (succinylcholine)
Normal neuromuscular function → mechanism of action	Compete with ACh at the nicotinic end-plate receptor	Agonist: depolarizes and holds the end plate (phase I), then desensitizes (phase II)
Skeletal muscle paralysis	Onset: rocuronium fastest (60–120 s)	Fasciculations, then paralysis < 90 s; lasts < 10 min
Neuromuscular response to cholinesterase inhibitors	Reversed (neostigmine); sugammadex for rocuronium/vecuronium	Phase I block augmented, not reversed
Cardiovascular effects	Vecuronium, cisatracurium, rocuronium: minimal; histamine release with atracurium/tubocurarine	Arrhythmias; bradycardia with a repeat dose
Neuromuscular blockade enhanced by	Volatile anesthetics (isoflurane most), aminoglycosides, myasthenia gravis, old age	Plasma cholinesterase variants (dibucaine number)
Key dangers	Residual block → hypoventilation	Hyperkalemia (burns, denervation); MH trigger; ↑ intraocular and intragastric pressure

How to tell them apart. Sources: p. 500; p. 503; p. 504; p. 507; p. 508; p. 509

WHY IT HAPPENS

- Why is dantrolene the malignant hyperthermia drug? MH is runaway calcium release through RyR1 in muscle; a CNS drug can't reach that — dantrolene blocks RyR1 itself. S34; S35; SG U2-R11
- Why does stopping intrathecal baclofen go badly? The spinal cord loses its GABA-B brake all at once: seizures, hyperthermia, and rebound spasticity. S27; S28

MNEMONIC (MEMORY AID)

- **Cyclobenzaprine looks like a tricyclic, acts like one:** sleepy, dry, can't pee, confused — and serotonin syndrome with SSRIs. S31; S32

HIGH-YIELD

- Central (cyclobenzaprine, tizanidine, baclofen) vs. peripheral (dantrolene — muscle; botulinum — junction). S23; S24; SG U2-R11
- Dantrolene blocks RyR1 SR Ca^{2+} release → malignant hyperthermia. S34; S35; SG U2-R11
- Baclofen: GABA-B; never stop intrathecal abruptly. S27; S28
- Tizanidine: α2; LFTs; no ciprofloxacin/fluvoxamine. S29; S30

- Cyclobenzaprine: anticholinergic, Beers list, 2–3 weeks. S31; S32

3 PART 3 · DRUG CLASS

Bone Pharmacology & Osteoporosis

SO3 SO4 IO2.1 IO2.2 IO2.3 IO2.4 IO2.5 IO2.6

Class

- Bone pharmacology is about **balancing bone formation and bone resorption**. S39; S67
- **Osteoblasts** build new bone matrix (promoted by several osteoporosis therapies); **osteoclasts** resorb existing bone (suppressed by most osteoporosis therapies). S40
- Nearly every osteoporosis drug shifts this balance one way or the other — slowing osteoclasts or stimulating osteoblasts. S40 (notes)
- **RANKL/OPG:** osteoblasts or stromal cells express RANKL → RANKL binds RANK on osteoclast precursors → osteoclast differentiation and activation → bone resorption releases calcium. S41
- **OPG** is a decoy receptor that blocks RANKL. Estrogen increases OPG; denosumab mimics OPG. Sclerostin inhibits Wnt signaling in osteoblasts. S41
- **PTH:** intermittent dosing favors formation; continuous exposure raises RANKL and resorption. S41
- **Calcium homeostasis:** PTH raises serum calcium (mobilizes bone calcium when levels drop); vitamin D increases intestinal calcium absorption; calcitonin lowers serum calcium and opposes PTH; calcium is the substrate being regulated (intake, bone storage, excretion). S42
- **PTH feedback loop:** falling serum calcium → PTH → raises calcium via bone resorption, renal reabsorption, and (via vitamin D activation) intestinal absorption; restored calcium inhibits further PTH. S43; SG U2-R12

Prototype + other drugs named

- Alendronate — oral bisphosphonate (first-line). S47
- Denosumab — RANKL monoclonal antibody. S54
- Teriparatide — PTH analog (anabolic). S54
- Raloxifene — SERM; romosozumab — sclerostin antibody. S54

- **Risk factors:** aging; estrogen deficiency (e.g., postmenopausal status); **chronic glucocorticoid use**; low calcium or vitamin D intake; low physical activity. S46
- **TB** About 98% of the body's calcium is in bone; PTH, FGF23, and 1,25-dihydroxyvitamin D are the principal regulators. p. 811; p. 812
- **TB** Excess endogenous PTH nets bone resorption, but low intermittent doses build bone first — the basis for teriparatide. p. 815
- **TB** Glucocorticoids block vitamin D–driven gut calcium transport, raise renal calcium loss, block formation, and initially stimulate resorption. p. 818

Mechanism of action

- Bisphosphonates — **mechanism: inhibit osteoclast-mediated bone resorption.** They bind hydroxyapatite at resorption sites → osteoclasts take them up → nitrogen-containing agents inhibit **FPP synthase** (mevalonate pathway) → loss of prenylation of small GTPases → **osteoclast apoptosis**. S47; S48
- **Denosumab:** monoclonal antibody, **RANKL inhibitor** — blocks osteoclast formation. S54; S55; SG U2-M
- **Teriparatide** (PTH 1-34, daily injection) / **abaloparatide:** PTH-analog "anabolic" agents that stimulate osteoblasts and new bone formation. S54; S55; SG U2-M
- **Romosozumab:** sclerostin antibody that both builds bone and limits resorption — increases formation, decreases resorption → BMD up, fracture risk down. S54; S55
- **SERMs** — selective estrogen receptor modulators (raloxifene): estrogen-like effect on bone — antiresorptive, like calcitonin. S54; S55
- **Glucocorticoid-induced osteoporosis:** glucocorticoids increase osteoclast-mediated resorption and suppress osteoblast-driven formation. S71
- **TB** Amino-bisphosphonates (alendronate, risedronate) inhibit farnesyl

Indications

- **Bisphosphonates are first-line for most patients with osteoporosis.** S47
- Higher-risk patients may need an **anabolic agent first**. S59
- Raloxifene also reduces breast cancer risk. S54; S55
- **TB** Raloxifene protects against spine fractures but not hip fractures; bisphosphonates, denosumab, and teriparatide protect both. p. 827
- **TB** Vitamin D (800 IU/d) with calcium improves bone density, reduces falls, and prevents fractures — the base under every osteoporosis drug. p. 827
- **TB** Calcitonin is less effective (spine fractures only); bisphosphonates and calcitonin are first-line for Paget disease. p. 828; p. 829

pyrophosphate synthase in the mevalonate pathway. p. 820

- TB Denosumab does not kill osteoclasts and does not deposit in bone, so its effect reverses when stopped. p. 820

Adverse effects (boxed warnings first)

- **Romosozumab: boxed warning for MI, stroke, and cardiovascular death.** S55
- TB Romosozumab: black box — contraindicated after MI or stroke within the past year. p. 820
- **Bisphosphonates:** esophageal irritation; hypocalcemia; osteonecrosis of the jaw (rare); atypical femoral fractures (rare, with long-term use). S48; S52
- **IV zoledronic acid:** acute-phase reaction (flu-like), mitigated with acetaminophen. S53
- **Raloxifene:** raises VTE risk; hot flashes. S55
- **Denosumab discontinuation causes rebound vertebral fractures** — transition to a bisphosphonate. S55
- TB Osteonecrosis of the jaw is rare at osteoporosis doses (~1/100,000 patient-years), more common with high-dose IV zoledronate for cancer. p. 820
- TB Denosumab: possible infection risk, transient hypocalcemia (CKD, vitamin D deficiency), and rebound resorption if interrupted. p. 820

Contraindications

- Bisphosphonates: contraindicated with severe renal impairment (CrCl below 30–35 mL/min) — some are avoided in significant renal impairment. S48; S59
- Oral bisphosphonates: avoid in esophageal stricture, achalasia, or inability to sit upright. S53
- TB Unlike bisphosphonates, denosumab can be used in advanced renal disease (not renally cleared) — watch calcium. p. 820

Drug interactions

- Food, calcium, coffee, and antacids reduce oral bisphosphonate absorption further. S53

Pharmacokinetics

- Bisphosphonates have a long skeletal half-life — the effect persists after stopping. S48
- Poor oral absorption (under 1%) → take fasting with plain water. S53
- **Dosing:** alendronate, risedronate — oral, typically weekly; ibandronate — oral, monthly; zoledronic acid — IV, as infrequently as once yearly. S47; S49

- Teriparatide is limited to 2 years lifetime, then follow with an antiresorptive. S55
- **TB** Bisphosphonates: less than 10% of an oral dose is absorbed; about half goes to bone, the rest is excreted unchanged in urine. Drug bound in bone is retained for months to years — unlike denosumab, whose effect fades when stopped. p. 819
- **TB** Denosumab 60 mg SC every 6 months; teriparatide 20 mcg SC daily; romosozumab 210 mg SC monthly for one year, then an antiresorptive. p. 819; p. 827

Monitoring

- **Oral counseling:** take with a **full glass of water**, first thing in the morning, on an empty stomach well before food, and **remain upright** (sitting or standing) for the specified period after dosing (30 minutes). S52; S53; SG U2-R13
- Ensure calcium and vitamin D repletion before starting; correct hypocalcemia first. S48; S53
- Consider a **drug holiday after 3–5 years** of oral therapy (osteonecrosis of the jaw, atypical femur fracture with long-term use). S48
- **TB** Reassess after 5 years of bisphosphonate (3 for zoledronate); a drug holiday if fracture risk allows. p. 820

CLINICAL PEARL / WHY

- The “remain upright” point is extremely high-yield: esophageal injury is largely preventable with correct administration. S52 (notes); S53
- **Can’t swallow pills?** Yearly IV zoledronic acid or a subcutaneous injectable (e.g., denosumab) avoids the oral route. S66; S59
- Dosing frequency is a major real-world driver of which bisphosphonate a patient actually takes consistently. S49 (notes)
- Teriparatide, abaloparatide, and romosozumab are the **anabolic exception** — they actively build new bone; most other osteoporosis drugs just slow resorption. S54 (notes); S56
- Glucocorticoid-induced osteoporosis is especially high-yield — it connects to the corticosteroid use for OA/RA in Session 1. S46 (notes)
- **Choosing therapy:** fracture risk, renal function, previous fracture history, glucocorticoid exposure, adherence (weekly oral vs. yearly IV vs. injectable), and route preference. This checklist is a direct echo of Lecture 1’s clinical decision-making framework. S59; S59 (notes)

- Name clues: **-dronate** bisphosphonates (zoledronic acid = zoledronate); **-sumab** bone monoclonal antibodies (denosumab RANKL, romosozumab sclerostin); **-fene** SERM (raloxifene); **-tide** PTH analogs (teriparatide, abaloparatide); calcitonin has no shared stem (hormone). S62

Step by step

RANKL → osteoclast S41

Osteoblast/stromal cell expresses RANKL →

RANKL binds RANK on osteoclast precursors →

Osteoclast differentiation and activation → Bone resorption releases calcium →

OPG (decoy) and denosumab block RANKL

How bisphosphonates kill osteoclasts S48

Bind hydroxyapatite at resorption sites → Osteoclast takes the drug up →

N-bisphosphonates inhibit FPP synthase → Loss of GTPase prenylation →

Osteoclast apoptosis → less resorption

Taking an oral bisphosphonate S52; S53; SG U2-R13

First thing in the morning, fasting → Full glass of plain water →

Stay upright 30 minutes → No food, calcium, coffee, or antacids before then →

Less esophageal irritation

Figure 6 · Calcium homeostasis: the PTH feedback loop. Original illustrative diagram (not from Katzung): ↓ serum calcium → parathyroid gland releases PTH → bone (↑ osteoclast resorption), kidney (↑ Ca^{2+} reabsorption, activates vitamin D), intestine (↑ Ca^{2+} absorption via vitamin D) → ↑ serum calcium (restored). Negative feedback restores calcium, which inhibits further PTH release; calcitonin opposes PTH (lowers serum calcium when it is elevated). This loop is the physiologic backdrop for every osteoporosis drug. S43; S43 (notes)

Figure 7 · Bisphosphonate dosing frequency. Original illustrative comparison (illustrative dosing-frequency values, not from Katzung): alendronate/risedronate (oral, weekly) 52 doses per year; ibandronate (oral, monthly) 12 per year; zoledronic acid (IV, yearly) 1 per year. Dosing frequency drives adherence. S49; S49 (notes)

Figure 8 · How osteoporosis drugs shift the remodeling balance. Original illustrative diagram (not from Katzung): a bone remodeling balance — bone resorption (osteoclasts) is inhibited by bisphosphonates, denosumab, SERMs (raloxifene), and calcitonin; bone formation (osteoblasts) is stimulated by teriparatide, abaloparatide, and romosozumab. The capstone bone figure: every drug lands on one side or the other. S56; S56 (notes)

Osteoporosis drugs at a glance

Drug	Side of the balance	Route · schedule	Key safety point
Alendronate, risedronate	Antiresorptive (osteoclast apoptosis)	Oral, weekly	Full glass of water, upright 30 min, fasting; esophagitis
Ibandronate	Antiresorptive	Oral, monthly	Same oral rules
Zoledronic acid	Antiresorptive	IV, yearly	Flu-like acute-phase reaction; avoid if CrCl < 30–35
Denosumab	Antiresorptive (RANKL antibody, mimics OPG)	Injection	Rebound vertebral fractures if stopped → bisphosphonate
Raloxifene (SERM)	Antiresorptive (estrogen-like on bone)	Oral	VTE; hot flashes; may ↓ breast cancer risk
Calcitonin	Antiresorptive (opposes PTH)	Not covered in materials	Not covered in materials
Teriparatide, abaloparatide	**Anabolic** (PTH analog)	Daily injection	2-year lifetime limit, then antiresorptive
Romosozumab	**Anabolic + antiresorptive** (sclerostin antibody)	Injection	**Boxed: MI, stroke, CV death**

How to tell them apart. Sources: S47–S56

Oral bisphosphonate rules — and why

Rule	Why
Fasting, first thing in the morning, plain water	Absorption is under 1%; food, calcium, coffee and antacids cut it further
Full glass of water	Washes the tablet past the esophagus
Stay upright 30 minutes	Lowers esophageal contact time and irritation
Not with stricture, achalasia, or inability to sit up	Tablet would sit in the esophagus
Correct calcium and vitamin D first	Shutting down resorption can drop calcium

How to tell them apart. Sources: S52; S53; SG U2-R13

Name clues: bone

Name clue	Class	Examples
-dronate	Bisphosphonates (inhibit osteoclast resorption)	Alendronate, risedronate, ibandronate; zoledronic acid (zoledronate)
-sumab	Monoclonal antibodies (-mab) targeting bone	Denosumab (RANKL), romosozumab (sclerostin)
-fene	Selective estrogen receptor modulator (SERM)	Raloxifene
-tide (peptide)	PTH analogs (anabolic)	Teriparatide, abaloparatide
No shared stem	Calcitonin (hormone)	Calcitonin

How to tell them apart. Sources: S62

Osteoporosis drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Alendronate (Fosamax)	**Bisphosphonate**	Tablets 5/10/35/70 mg; effervescent tablet; solution	70 mg PO once weekly (or 10 mg daily) fasting with 8 oz plain water, upright 30 minutes	Postmenopausal osteoporosis, glucocorticoid-induced osteoporosis, Paget disease

Drug	Class	Formulation	Adult dose	Indications
Zoledronic acid (Reclast, Zometa)	**Bisphosphonate (IV)**	IV 5 mg/100 mL (Reclast); 4 mg (Zometa)	5 mg IV once yearly over at least 15 minutes (osteoporosis); once every 2 years for prevention	Osteoporosis, Paget disease, hypercalcemia of malignancy, skeletal metastases
Denosumab (Prolia, Xgeva)	**RANKL inhibitor (monoclonal antibody)**	SC prefilled syringe 60 mg/mL (Prolia); 120 mg (Xgeva)	60 mg SC every 6 months (Prolia)	Postmenopausal osteoporosis, bone loss with androgen deprivation, skeletal events
Teriparatide (Forteo)	**Anabolic PTH(1-34) analog**	Prefilled pen 20 mcg/dose	20 mcg SC once daily in thigh or abdomen; lifetime use limited to 2 years	Severe osteoporosis, high fracture risk, glucocorticoid-induced osteoporosis
Raloxifene (Evista)	**SERM**	Tablets 60 mg	60 mg PO once daily	Postmenopausal osteoporosis; reduction of invasive breast cancer risk

How to tell them apart. Sources: S84; S85; S86; S87; S88

Osteoporosis drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Alendronate (Fosamax)	Esophagitis, dyspepsia, musculoskeletal pain, hypocalcemia	Esophageal stricture or achalasia, inability to sit upright, hypocalcemia, CrCl below 35 mL/min	Esophageal injury, hypocalcemia; osteonecrosis of the jaw and atypical femur fracture with long-term use
Zoledronic acid (Reclast, Zometa)	Flu-like acute-phase reaction, hypocalcemia, renal injury, osteonecrosis of the jaw	CrCl below 35 mL/min, uncorrected hypocalcemia, pregnancy	Acute kidney injury and hypocalcemia; hydrate; replete calcium and vitamin D first
Denosumab (Prolia, Xgeva)	Back and joint pain, hypocalcemia, infection, osteonecrosis of the jaw, atypical femur fracture	Hypocalcemia, pregnancy; severe hypocalcemia boxed warning in advanced CKD	Rebound multiple vertebral fractures if stopped or doses missed; transition to a bisphosphonate
Teriparatide (Forteo)	Orthostatic hypotension, leg cramps, nausea, hypercalcemia, dizziness	Paget disease, prior skeletal radiation, bone metastases, hypercalcemia, open epiphyses	Hypercalcemia and orthostatic hypotension; follow with an antiresorptive to keep gains
Raloxifene (Evista)	Hot flashes, leg cramps, venous thromboembolism, peripheral edema	History of VTE, pregnancy, breastfeeding; stroke risk (caution)	Boxed warning for VTE and death from stroke; stop before prolonged immobilization

How to tell them apart. Sources: S84; S85; S86; S87; S88

Hormones and drugs in bone mineral homeostasis (textbook)

Regulator / drug	What it does	Clinical use or point
Parathyroid hormone (PTH)	↑ serum calcium, ↓ phosphate; ↑ $1,25(OH)_2D$; RANKL via osteoblasts	Intermittent PTH 1-34 (teriparatide) is anabolic; rhPTH 1-84 for hypoparathyroidism
Vitamin D (1,25-dihydroxyvitamin D)	↑ gut calcium and phosphate absorption	Vitamin D preparations: D3 preferred; calcitriol when speed matters
Fibroblast growth factor 23	↓ phosphate reabsorption and $1,25(OH)_2D$	Burosumab (anti-FGF23) for X-linked hypophosphatemia
Calcitonin (secondary regulator)	Inhibits osteoclasts; ↓ calcium and phosphate	Salmon form lasts 40–50 min; Paget, hypercalcemia, osteoporosis (weaker)
Glucocorticoids	Net bone loss	Common cause of adult osteoporosis; treat sarcoid/vitamin D hypercalcemia
Estrogens	Reduce PTH-driven resorption	Prevent postmenopausal loss; long-term use out of favor → SERMs
Sclerostin antibodies (romosozumab)	Unblock Wnt → osteoblasts; ↓ resorption	1 year, then antiresorptive; MI/stroke boxed warning
Calcimimetics (cinacalcet, etelcalcetide)	Activate the calcium-sensing receptor → ↓ PTH	Secondary hyperparathyroidism in CKD; cinacalcet 30 → 90 mg
Thiazide diuretics	↓ renal calcium excretion	Idiopathic hypercalciuria stones; avoid loop diuretics there
Fluoride	Stabilizes hydroxyapatite	Dental caries; not FDA approved for osteoporosis (fractures not reduced)
Strontium ranelate	↓ resorption, ↑ formation	Europe only; not approved in the USA

How to tell them apart. Sources: p. 813; p. 815; p. 816; p. 817; p. 818; p. 819; p. 820; p. 821; p. 824; p. 825

Calcium and phosphate disorders (textbook, clinical pharmacology)

Disorder	Drug approach	Pearl
Hypercalcemia	Saline diuresis 500–1000 mL/h, then furosemide; pamidronate 60–90 mg or zoledronate 4 mg IV; calcitonin	Calcitonin acts in 4–6 h; gallium nitrate seldom used; glucocorticoids for sarcoid, vitamin D, myeloma
Hypocalcemia	IV 10% calcium gluconate slowly; oral 1000–1500 mg elemental/day; calcitriol 0.25–1 mcg	Calcium carbonate 40% calcium (with meals if achlorhydric) vs. citrate 21%; check magnesium
Hyperphosphatemia	Diet + binders: sevelamer, lanthanum, calcium	Aluminum binders sparingly (bone disease)
Hypophosphatemia	Oral phosphate	Long term: weakness, osteomalacia

Disorder	Drug approach	Pearl
Primary hyperparathyroidism	Surgery; cinacalcet if not a candidate	Rule out familial hypocalciuric hypercalcemia
Hypoparathyroidism	Calcitriol + calcium; rhPTH 1-84	Hypercalciuria limits standard therapy
Nutritional vitamin D deficiency or insufficiency	D2 or D3 1000–4000 IU/d or 50,000 IU/week	25(OH)D > 12 ng/mL prevents rickets; 20–30 ng/mL target
Chronic kidney disease	Calcitriol, doxercalciferol, paricalcitol; binders	Keep calcium × phosphate < 55
Intestinal osteodystrophy (malabsorption)	High-dose vitamin D 25,000–50,000 IU	Osteoporosis + osteomalacia
Idiopathic hypercalciuria	Hydrochlorothiazide up to 50 mg twice daily	Not loop diuretics; allopurinol an alternative
Paget disease of bone	Bisphosphonates (higher doses), calcitonin	Follow alkaline phosphatase
Enteric oxaluria	Calcium carbonate 1–2 g/day	Calcium binds gut oxalate

How to tell them apart. Sources: p. 822; p. 823; p. 824; p. 825; p. 826; p. 829

SLIDE DISCREPANCY

- **Oral bisphosphonate absorption**
- Slides: Poor oral absorption (under 1%). S53
- Textbook: Less than 10% of an oral dose is absorbed. p. 819

WHY IT HAPPENS

- Why does stopping denosumab cause fractures? It blocks osteoclast formation without killing them; stop it and resorption rebounds — so hand off to a bisphosphonate. S55
- Why the strict bisphosphonate rules? Under 1% is absorbed, and a tablet lodged in the esophagus burns it — fasting, plain water, upright. S53; SG U2-R13

MNEMONIC (MEMORY AID)

- **PTH pulls from three: Bone, Kidney, Gut.** Intermittent PTH builds; continuous PTH breaks down. S41; S43; SG U2-R12

HIGH-YIELD

- Bone pharmacology is fundamentally about the balance between osteoblast and osteoclast activity. S67; S40
- PTH: falling calcium → bone resorption, renal reabsorption, vitamin D–mediated gut absorption. S43; SG U2-R12

- Bisphosphonates first-line; full glass of water, upright 30 minutes. S47; S52; SG U2-R13
- Anabolic: teriparatide/abaloparatide (2-year limit), romosozumab (boxed CV). S54; S55
- Denosumab = RANKL antibody; stopping → rebound vertebral fractures. S54; S55
- Glucocorticoids: ↑ resorption, ↓ formation → osteoporosis. S46; S71

4 PART 4 · TOPIC

Putting It All Together: Musculoskeletal Toolkit

SO5 IO2.3 IO2.4

Key definitions

- Applying the whole musculoskeletal pharmacology toolkit: for each patient, name a first-line drug class AND the single biggest safety concern to check before prescribing it. S63; S64 (notes)

How it works

- **1.** Pain and inflammation are related but distinct — treat them accordingly. S67
- **2.** NSAIDs are effective for pain and inflammation, but carry real GI, renal, and cardiovascular risk. S67
- **3.** DMARDs modify the underlying disease process in RA — they do more than mask symptoms. S67
- **4.** Acute gout treatment and chronic urate-lowering therapy are different goals, started at different times. S67
- **5.** Bone pharmacology is fundamentally about the balance between osteoblast and osteoclast activity. S67
- Every mechanism here — COX inhibition, CYP interactions, renal dosing, pharmacogenomics — is Lecture 1's toolkit applied to one organ system. S67 (notes)

Drugs named as examples

- **Recall Q1:** acetaminophen is not a reliable first-line choice for an inflammatory joint flare — it has minimal peripheral anti-inflammatory effect; it treats pain and fever, not inflammation itself. S69; S71
- **Recall Q2:** a 70-year-old with CKD and heart failure needing OA pain control — acetaminophen or a topical NSAID avoids the systemic renal/CV risk of oral NSAIDs. S69; S71
- **Recall Q3:** methotrexate inhibits dihydrofolate reductase; allopurinol inhibits xanthine oxidase; colchicine disrupts microtubules; baclofen is a GABA-B agonist. S69; S71

- **Recall Q4:** fever on a biologic DMARD — immunosuppression is the shared major risk across all DMARDs; treat a fever as a possible infection until proven otherwise. S70; S71
- **Recall Q5:** probenecid is a uricosuric that inhibits renal tubular reabsorption of uric acid. Colchicine does not lower uric acid; febuxostat (not probenecid) is a xanthine oxidase inhibitor; ULT is deferred during a flare. S70; S71
- **Recall Q6:** glucocorticoids increase osteoclast-mediated bone resorption and suppress osteoblast-driven bone formation — shifting the balance toward bone loss. S70; S71

CLINICAL

- **Patient A — acute gout flare, first presentation:** colchicine, NSAID, or corticosteroid; check renal/hepatic function and CYP3A4/P-gp inhibitors (colchicine). No ULT yet. S64; S9; S11
- **Patient B — chronic RA, inadequately controlled:** escalate DMARD therapy (biologic); check infection risk — TB and hepatitis screening first. S64; S70; S71
- **Patient C — osteoarthritis + chronic kidney disease:** acetaminophen or a topical NSAID; avoid oral NSAIDs (renal risk). S64; S69; S71
- **Patient D — postmenopausal osteoporosis, high fracture risk:** consider an anabolic agent first; check its limits (teriparatide 2 years; romosozumab CV boxed warning). S64; S59; S55

TRAP

- Colchicine treats the flare but **does not lower uric acid**; probenecid is **not** a xanthine oxidase inhibitor. S71
- Baclofen is a GABA-B agonist — not a calcium channel blocker in skeletal muscle (that idea belongs to dantrolene's RyR1 site). S71; S35
- OA in the CKD + anticoagulant + hypertension patient: oral NSAIDs carry the highest combined risk — CKD (renal/AKI risk), anticoagulant (bleeding risk), hypertension (BP/CV risk); acetaminophen or a topical NSAID is generally safer first-line. SG U2-S
- When that patient flares with gout: an oral NSAID is still relatively contraindicated; a short corticosteroid course (oral or intra-articular) is often preferred, colchicine needs renal dose adjustment. Defer allopurinol/febuxostat until the flare resolves. SG U2-S

Study-guide mechanism drill: for each drug, write its primary mechanism of action

Drug	Mechanism (your answer)
Methotrexate	Inhibits dihydrofolate reductase (Session 1)

Drug	Mechanism (your answer)
Colchicine	Disrupts neutrophil microtubule formation
Dantrolene	Decreases SR calcium release
Baclofen	GABA-B agonist (spinal cord)
Teriparatide	PTH analog — stimulates osteoblast bone formation
Denosumab	RANKL inhibitor — blocks osteoclast formation
Abatacept	Blocks T-cell costimulation (Session 1)
Spironolactone (as used for hirsutism/acne)	Blocks the androgen receptor (antiandrogen effect)

How to tell them apart. Sources: SG U2-M

Which drug would you choose? (S64)

Patient	First-line class	Biggest safety check
A · Acute gout flare, first presentation	Colchicine, NSAID, or corticosteroid	Defer ULT; colchicine renal/hepatic dosing and CYP3A4/P-gp inhibitors
B · Chronic RA, inadequately controlled	Add/escalate DMARD (biologic)	Infection: TB and hepatitis screening; fever = infection until proven otherwise
C · Osteoarthritis + CKD	Acetaminophen or topical NSAID	Avoid oral NSAIDs (AKI)
D · Postmenopausal osteoporosis, high fracture risk	Anabolic agent first, then antiresorptive	Teriparatide 2-year limit; romosozumab boxed CV warning; renal function

How to tell them apart. Sources: S64; S59; S55; S69–S71

Supporting drugs at a glance (listed for context; not profiled in detail)

Drug(s)	Why it matters
NSAIDs and corticosteroids	Alternative acute gout flare treatment
Azathioprine and mercaptopurine	Allopurinol raises levels; reduce dose
Ciprofloxacin and fluvoxamine	CYP1A2 inhibitors; avoid with tizanidine
Clarithromycin	CYP3A4/P-gp inhibitor raises colchicine toxicity
Thiazide and loop diuretics	Raise uric acid levels
Opioids, benzodiazepines and alcohol	Additive sedation with relaxants
Volatile anesthetics and succinylcholine	Triggers of malignant hyperthermia

Drug(s)	Why it matters
Calcium and vitamin D	Replete before and during osteoporosis therapy
Risedronate, ibandronate, romosozumab, abaloparatide, calcitonin	Other osteoporosis agents

How to tell them apart. Sources: S89; S89 (notes)

MNEMONIC (MEMORY AID)

- Every case, two answers: **the class** and **the one thing to check before you write it**. S64 (notes)

HIGH-YIELD

- Five takeaways: pain ≠ inflammation; NSAIDs = GI/renal/CV risk; DMARDs change RA; gout = acute vs. chronic; bone = osteoblast vs. osteoclast. S67

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
Gout chain	Purine → xanthine → uric acid (xanthine oxidase) → MSU crystals → neutrophil flare S5; S6
Raise uric acid	Purine/fructose diet, beer, thiazide/loop diuretics, CKD S5
Flare drugs	Colchicine, NSAIDs, corticosteroids S9
Urate-lowering drugs	Allopurinol, febuxostat (↓ production); probenecid (↑ excretion) S9; SG U2-R9
ULT in a flare?	Never start or change — it can prolong the flare S9; SG U2-R10
ULT prophylaxis	Colchicine or NSAID for 3–6 months S13
Colchicine	Microtubules · best within 24 h · diarrhea · CYP3A4/P-gp S10; S11
Allopurinol	XO inhibitor (oxypurinol) · first-line · HLA-B*58:01 · ↓ dose renal, azathioprine S12; S13
Febuxostat	XO inhibitor · boxed CV death S13
Probenecid	URAT1 block · hydrate · no uric acid stones/renal impairment S14; S15
Pegloticase, rasburicase	Uricase → allantoin · IV · no G6PD deficiency S15
Spasticity vs. rigidity	Velocity-dependent (UMN) vs. constant tone S21
Central relaxants	Cyclobenzaprine, tizanidine, baclofen (+ carisoprodol, methocarbamol, metaxalone, chlorzoxazone) S23
Peripheral relaxants	Dantrolene (muscle), botulinum toxin (junction) S23
Baclofen	GABA-B · MS/spinal cord spasticity · intrathecal · withdrawal seizures, hyperthermia S27; S28
Tizanidine	α2 agonist · hypotension, LFTs · no ciprofloxacin/fluvoxamine S29; S30
Cyclobenzaprine	Brainstem 5-HT2 · TCA-like anticholinergic · 2–3 weeks · Beers S31; S32
Carisoprodol	→ meprobamate · Schedule IV S33
Dantrolene	RyR1 · malignant hyperthermia, NMS · hepatotoxic S34; S35
Malignant hyperthermia triggers	Volatile anesthetics, succinylcholine S35
Osteoporosis risk	Age, estrogen loss, glucocorticoids, low Ca/vitamin D, inactivity S46

Prompt	Answer
PTH loop	↓ Ca → PTH → bone resorption, renal reabsorption, vitamin D gut absorption S43; SG U2-R12
Intermittent vs. continuous PTH	Formation vs. resorption (↑ RANKL) S41
Bisphosphonate schedules	Alendronate/risedronate weekly · ibandronate monthly · zoledronic acid yearly IV S47; S49
Bisphosphonate counseling	Full glass of water, fasting, upright 30 min S52; S53; SG U2-R13
Bisphosphonate limits	CrCl < 30–35 · ONJ, atypical femur fracture · holiday after 3–5 y S48
Denosumab	RANKL antibody · stopping → rebound vertebral fractures S54; S55
Teriparatide	PTH 1-34 · anabolic · 2 years lifetime S55; SG U2-M
Romosozumab	Sclerostin antibody · builds + limits resorption · boxed MI/stroke/CV death S54; S55
Raloxifene	SERM · VTE, hot flashes · ↓ breast cancer risk S54; S55
Name stems	-ostat XO · -ase uricase · -curonium NMB · -dronate · -sumab · -fene SERM · -tide PTH S19; S38; S62
Allopurinol dosing	Start 100 mg/d (≤50 in CKD); titrate to uric acid <6 mg/dL; max 800 p. 695
Allopurinol hypersensitivity	2–4 weeks in; stop at first rash; screen HLA-B 5801 in Han Chinese, Korean, Thai p. 696
Colchicine flare dose	1.2 mg, then 0.6 mg 1 h later p. 696
Gout steroid option	Prednisone 30–50 mg/d × 1–2 days, taper 7–10 days; good in CKD p. 698
Pegloticase dose	8 mg IV every 2 weeks; premedicate p. 698
Probenecid	Start 2–3 weeks after an attack; urine pH > 6.0; ↓ penicillin secretion p. 698
Dantrolene in MH	1 mg/kg IV, repeat; RyR1 (heart uses RyR2) p. 513
Succinylcholine dangers	Hyperkalemia (burns, denervation, trauma); MH trigger p. 508
Sugammadex	Reverses rocuronium/vecuronium; backup contraception 7 days p. 509; p. 510
Bisphosphonate doses	Alendronate 70 mg/wk · risedronate 35 mg/wk · ibandronate 150 mg/mo · zoledronate 5 mg/yr p. 818
Teriparatide · abaloparatide	20 mcg · 80 mcg SC daily; 2 years p. 827
Raloxifene	Spine, not hip; VTE; no hot-flush relief p. 827
Vitamin D	600 IU/d adults; deficiency 1000–4000 IU/d or 50,000 IU/week p. 825

Self-test

First, the lecture's own 15 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Lecture check 1 · S7-8

In Figure 4, probenecid and allopurinol both lower serum uric acid, but through different mechanisms. What is the key difference?

- **A** Probenecid increases renal excretion of uric acid; allopurinol decreases uric acid production
- **B** Both drugs block xanthine oxidase
- **C** Probenecid decreases uric acid production; allopurinol increases renal excretion
- **D** Neither drug affects serum uric acid levels

ANSWER: A

Figure 4 shows allopurinol (and febuxostat) acting as xanthine oxidase inhibitors that reduce uric acid production, while probenecid acts at the kidney to increase renal excretion of uric acid that has already been produced — two different strategies for the same goal of lowering serum uric acid. S7-8

Lecture check 2 · S17-18

A patient presents with an acute gout flare. Which action is most appropriate at this visit?

- **A** Treat the acute flare (colchicine, NSAID, or corticosteroid) and defer urate-lowering therapy
- **B** Start allopurinol immediately to lower uric acid
- **C** Start febuxostat and allopurinol together
- **D** No treatment is needed until uric acid is measured

ANSWER: A

Starting or changing urate-lowering therapy during an acute flare can worsen or prolong it. Treat the flare first; urate-lowering therapy is started or adjusted only once it resolves. S17-18

Lecture check 3 · S25-26

In Figure 5, which agent is the odd one out, acting directly on the skeletal muscle fiber rather than within the central nervous system?

- **A** Cyclobenzaprine
- **B** Tizanidine
- **C** Baclofen
- **D** Dantrolene

ANSWER: D

Dantrolene decreases calcium release from the skeletal muscle sarcoplasmic reticulum, acting directly on the muscle fiber — unlike cyclobenzaprine, tizanidine, and baclofen, which all act within the CNS to reduce excitatory output to motor neurons. S25-26

Lecture check 4 · S36-37

Which skeletal muscle relaxant acts directly on skeletal muscle itself (not the CNS), making it the agent of choice in malignant hyperthermia?

- **A** Baclofen
- **B** Dantrolene
- **C** Tizanidine
- **D** Cyclobenzaprine

ANSWER: B

Dantrolene’s peripheral mechanism — decreasing calcium release from the skeletal muscle sarcoplasmic reticulum — is unique among these agents and is exactly why it is the emergency treatment for malignant hyperthermia. S36-37

Lecture check 5 · S44-45

In Figure 6, PTH release raises serum calcium through three parallel actions. Which organ system is NOT one of the three direct targets of PTH shown in the diagram?

- **A** Bone
- **B** Kidney
- **C** Intestine (directly)
- **D** Lung

ANSWER: D

PTH acts directly on bone (↑ osteoclast resorption) and kidney (↑ calcium reabsorption, activates vitamin D); its effect on the intestine is indirect, mediated through activated vitamin D. The lung is not part of this feedback loop at all. S44-45

Lecture check 6 · S50-51

Based on Figure 7, which bisphosphonate requires the fewest doses per year, and by what route is it given?

- **A** Alendronate/risedronate, given orally
- **B** Ibandronate, given orally
- **C** All bisphosphonates require the same number of annual doses
- **D** Zoledronic acid, given intravenously

ANSWER: D

Figure 7 shows zoledronic acid dosed as infrequently as once yearly by IV infusion, versus weekly oral dosing for alendronate/risedronate and monthly oral dosing for ibandronate — an important consideration for patients with adherence challenges. S50-51

Lecture check 7 · S57-58

In Figure 8, teriparatide and romosozumab sit on the "bone formation" side of the balance, while alendronate and raloxifene sit on the "bone resorption" side. What does this grouping represent?

- **A** Teriparatide and romosozumab stimulate osteoblast activity, while alendronate and raloxifene inhibit osteoclast activity
- **B** All four drugs work through the identical mechanism
- **C** Teriparatide and romosozumab inhibit osteoclasts, while alendronate and raloxifene stimulate osteoblasts
- **D** This grouping reflects route of administration, not mechanism

ANSWER: A

The "anabolic" agents (teriparatide, abaloparatide, romosozumab) sit on the bone-formation side because they stimulate osteoblasts, while bisphosphonates (alendronate), denosumab, SERMs (raloxifene), and calcitonin sit on the bone-resorption side because they inhibit osteoclasts — two different strategies for shifting the same remodeling balance. S57-58

Lecture check 8 · S60-61

A patient is counseled to take their oral bisphosphonate with a full glass of water and remain upright for a specified period. What complication is this counseling primarily meant to prevent?

- **A** Hypocalcemia
- **B** Atypical femoral fracture
- **C** Esophageal irritation or injury
- **D** Osteonecrosis of the jaw

ANSWER: C

Taking the tablet with a full glass of water and staying upright reduces the time the pill is in contact with the esophageal mucosa, lowering the risk of esophageal irritation or injury. S60-61

Lecture check 9 · S65-66

A patient with osteoporosis has significant difficulty swallowing pills. Which administration route among osteoporosis therapies would best fit this patient?

- **A** Weekly oral alendronate
- **B** Daily oral risedronate
- **C** There is no alternative to oral bisphosphonates
- **D** Yearly IV zoledronic acid or a subcutaneous injectable agent

ANSWER: D

Route of administration and adherence are explicit factors in choosing osteoporosis therapy — IV zoledronic acid or an injectable agent (e.g., denosumab) avoids the swallowing difficulty that limits oral options. S65-66

Lecture check 10 · S69; S71

Why is acetaminophen not a reliable first-line choice for an inflammatory joint flare, despite being a good analgesic/antipyretic?

- **A** It strongly inhibits COX-2 only
- **B** It has minimal peripheral anti-inflammatory effect
- **C** It is contraindicated in all patients with joint pain
- **D** It is a DMARD and too strong for mild flares

ANSWER: B

Acetaminophen has minimal peripheral anti-inflammatory effect, unlike NSAIDs — it treats pain and fever, not inflammation itself. S69; S71

Lecture check 11 · S69; S71

A 70-year-old with CKD and heart failure needs osteoarthritis pain control. Considering systemic risk, which is generally the safest starting point?

- **A** Oral ibuprofen
- **B** Oral celecoxib
- **C** Acetaminophen or a topical NSAID
- **D** Ketorolac

ANSWER: C

Acetaminophen or a topical NSAID avoids the systemic renal/CV risk of oral NSAIDs in a patient with CKD and heart failure. S69; S71

Lecture check 12 · S69; S71

Which pairing correctly matches a drug with its primary mechanism?

- **A** Allopurinol — microtubule disruption
- **B** Colchicine — xanthine oxidase inhibition
- **C** Baclofen — calcium channel blockade in skeletal muscle
- **D** Methotrexate — dihydrofolate reductase inhibition

ANSWER: D

Methotrexate inhibits dihydrofolate reductase. Allopurinol inhibits xanthine oxidase; colchicine disrupts microtubules; baclofen is a GABA-B agonist. S69; S71

Lecture check 13 · S70; S71

A patient on a biologic DMARD develops a fever. What is the most important immediate concern?

- **A** Infection, given the drug's immunosuppressive effect
- **B** A drug allergy only
- **C** Osteonecrosis of the jaw
- **D** Hepatotoxicity from Phase II metabolism

ANSWER: A

Immunosuppression is the shared major risk across all DMARDs — a fever must be treated as a possible infection until proven otherwise. S70; S71

Lecture check 14 · S70; S71

Which statement about gout pharmacology is correct?

- **A** Colchicine lowers uric acid production
- **B** Acute flares should always be treated with urate-lowering therapy first
- **C** Probenecid lowers uric acid by inhibiting renal tubular reabsorption of urate
- **D** Febuxostat and probenecid are both xanthine oxidase inhibitors

ANSWER: C

Probenecid is a uricosuric agent that inhibits renal tubular reabsorption of uric acid, increasing its excretion. Colchicine does not lower uric acid; febuxostat (not probenecid) is a xanthine oxidase inhibitor; and urate-lowering therapy is deferred during an acute flare. S70; S71

Lecture check 15 · S70; S71

A patient with glucocorticoid-induced osteoporosis risk should be counseled that the risk is primarily related to which mechanism?

- **A** Increased osteoclast-mediated resorption and decreased bone formation
- **B** Increased osteoblast activity
- **C** Excess vitamin D
- **D** Increased calcitonin secretion

ANSWER: A

Glucocorticoids increase osteoclast-mediated bone resorption and suppress osteoblast-driven bone formation — shifting the remodeling balance toward bone loss. S70; S71

Original questions

Q1 · Mechanism · Gout: Acute Flares & Urate-Lowering Therapy

How does colchicine relieve an acute gout flare?

- **A** It disrupts microtubules, reducing neutrophil migration
- **B** It blocks phospholipase A_2 upstream of COX
- **C** It converts uric acid to allantoin in the joint
- **D** It blocks URAT1, increasing uric acid excretion
- **E** It inhibits xanthine oxidase, lowering uric acid

ANSWER: A

Colchicine binds tubulin and blocks microtubule polymerization, reducing neutrophil migration and NLRP3 activation (S10; S11; SG U2-M). S10; S11; SG U2-M

- **B.** That is the corticosteroid site (Session 1).
- **C.** That is uricase (S15).
- **D.** That is probenecid (S15).
- **E.** That is allopurinol/febuxostat; colchicine doesn't lower uric acid (S71).

Q2 · Clinical vignette · Gout: Acute Flares & Urate-Lowering Therapy

A patient on colchicine for a gout flare is prescribed clarithromycin for pneumonia. What is the concern?

- **A** Clarithromycin speeds colchicine clearance, so the flare returns
- **B** Clarithromycin triggers HLA-B*58:01 hypersensitivity
- **C** CYP3A4/P-gp inhibition raises colchicine to toxic levels
- **D** The pair raises uric acid and worsens the flare
- **E** The two together cause hemolysis in G6PD deficiency

ANSWER: C

CYP3A4 and P-gp inhibitors such as clarithromycin, cyclosporine, and strong CYP3A4 inhibitors raise colchicine toxicity (S11). S11

- **A.** It is an inhibitor; levels rise (S11).
- **B.** HLA-B*58:01 relates to allopurinol (S13).
- **D.** The taught interaction is colchicine toxicity (S11).
- **E.** Hemolysis in G6PD deficiency is the uricase warning (S15).

Q3 · Clinical vignette · Gout: Acute Flares & Urate-Lowering Therapy

A patient of Han Chinese descent with recurrent gout is about to start allopurinol. Which pre-treatment step does the lecture tie to Lecture 1's pharmacogenomics?

- **A** G6PD testing for hemolysis risk on the drug
- **B** HLA-B*58:01 testing for severe hypersensitivity risk
- **C** TPMT testing for bone marrow toxicity risk
- **D** CYP2D6 genotyping for poor metabolizer status
- **E** Baseline joint aspiration for urate crystals

ANSWER: B

The HLA-B*58:01 allele raises the risk of severe cutaneous/hypersensitivity reactions to allopurinol (S12; S13); the textbook recommends screening in Han Chinese, Korean, and Thai patients (p. 696). S12; S13; p. 696

- **A.** G6PD screening applies to pegloticase and rasburicase (S15).
- **C.** TPMT relates to azathioprine (Lecture 1).
- **D.** Not the allopurinol gene taught (S12).
- **E.** Not a pharmacogenomic step (S12).

Q4 · Clinical vignette · Gout: Acute Flares & Urate-Lowering Therapy

A patient with refractory tophaceous gout is being considered for pegloticase. Which test must come first?

- **A** HLA-B*58:01, because of skin reactions
- **B** Creatine kinase, because of rhabdomyolysis
- **C** A 24-hour urine urate, to rule out stones
- **D** Liver enzymes, because of hepatotoxicity
- **E** G6PD activity, because of hemolysis risk

ANSWER: E

Pegloticase and rasburicase are contraindicated in G6PD deficiency (hemolysis risk) (S15). S15

- **A.** HLA-B*58:01 is the allopurinol screen (S13).
- **B.** Not the taught pegloticase check (S15).
- **C.** Uric acid stones are a probenecid concern (S14).
- **D.** The lecture's pegloticase warning is G6PD and infusion reactions (S15).

Q5 · Clinical vignette · Gout: Acute Flares & Urate-Lowering Therapy

A patient with gout and CKD stage 3 is starting allopurinol. Which textbook starting dose fits?

- **A** Allopurinol is not used at all in CKD
- **B** 300 mg/d from day one to reach target fast
- **C** 100 mg/d, the same as normal kidney function
- **D** No more than 50 mg/d, titrated by 50 mg
- **E** 800 mg/d, the usual maximum dose

ANSWER: D

In chronic kidney disease the initial allopurinol dose should not exceed 50 mg/d, with 50-mg titrations (p. 695). p. 695

- **A.** It is used with a lower start (p. 695).
- **B.** High starting doses raise hypersensitivity risk (p. 696).
- **C.** 100 mg/d is the usual start without CKD (p. 695).
- **E.** 800 mg/d is the ceiling, not the start (p. 695).

Q6 · Clinical vignette · Skeletal Muscle Relaxants

During general anesthesia with a volatile agent and succinylcholine, a patient develops rising temperature, rigidity, and acidosis. Which drug is the emergency treatment, and where does it act?

- **A** Baclofen, activating GABA-B receptors in the cord
- **B** Cyclobenzaprine, acting in the brainstem
- **C** Dantrolene, blocking RyR1 calcium release in muscle
- **D** Botulinum toxin, blocking acetylcholine release
- **E** Tizanidine, activating central alpha-2 receptors

ANSWER: C

Malignant hyperthermia is triggered by volatile anesthetics and succinylcholine; dantrolene blocks RyR1 so less calcium leaves the sarcoplasmic reticulum (S34; S35; SG U2-R11). S34; S35; SG U2-R11

- **A.** Baclofen is a central spasticity drug, not the MH treatment (S27).
- **B.** Central agents don't reach the muscle defect (S24).
- **D.** Local, temporary; not an emergency drug (S33).
- **E.** Central agent; not the MH emergency drug (S29).

Q7 · Clinical vignette · Skeletal Muscle Relaxants

A patient with multiple sclerosis has an intrathecal baclofen pump. The pump runs dry over a holiday weekend. What should the team anticipate?

- **A** Serotonin syndrome from loss of 5-HT2 block
- **B** No change, since baclofen has a long half-life
- **C** Withdrawal: seizures, hyperthermia, rebound spasticity
- **D** Rebound hypertension from loss of α2 action
- **E** Malignant hyperthermia that needs dantrolene

ANSWER: C

Abrupt discontinuation, especially intrathecal, can cause a dangerous withdrawal syndrome with seizures, hyperthermia, and rebound spasticity (S27; S28). S27; S28

- **A.** Serotonin syndrome is a cyclobenzaprine interaction (S32).
- **B.** Withdrawal is the taught danger (S27).
- **D.** That fits tizanidine, not baclofen (S30).
- **E.** MH is triggered by anesthetics and succinylcholine (S35).

Q8 · Clinical vignette · Skeletal Muscle Relaxants

A patient taking tizanidine for spasticity is prescribed ciprofloxacin for a urinary infection. What is the problem?

- **A** Ciprofloxacin blocks RyR1 and adds weakness
- **B** The pair causes serotonin syndrome
- **C** Ciprofloxacin induces CYP1A2, so spasticity returns
- **D** Tizanidine stops ciprofloxacin being absorbed
- **E** CYP1A2 inhibition sharply raises tizanidine levels

ANSWER: E

Ciprofloxacin and fluvoxamine strongly raise tizanidine levels via CYP1A2 — the combination is contraindicated (S30). S30

- **A.** RyR1 is dantrolene's target (S35).
- **B.** Serotonin syndrome is cyclobenzaprine with SSRIs/SNRIs/tramadol (S32).
- **C.** Ciprofloxacin inhibits, not induces (S30).
- **D.** The taught interaction is raised tizanidine (S30).

Q9 · Clinical vignette · Skeletal Muscle Relaxants

A 78-year-old asks for cyclobenzaprine for back spasm. Which concern is most specific to this drug in an older adult?

- **A** Anticholinergic confusion and urinary retention
- **B** Hepatotoxicity requiring routine LFT monitoring
- **C** Seizures on abrupt withdrawal of a pump
- **D** Malignant hyperthermia during later anesthesia
- **E** Hemolysis in a patient with G6PD deficiency

ANSWER: A

Cyclobenzaprine is tricyclic-like: dry mouth, blurred vision, urinary retention, confusion — on the Beers list in older adults; short-term use only (S31; S32). S31; S32

- **B.** Routine LFTs are emphasized for tizanidine (S29).
- **C.** That is intrathecal baclofen (S28).
- **D.** MH is treated by dantrolene, not caused by cyclobenzaprine (S35).
- **E.** G6PD is the uricase warning (S15).

Q10 · Mechanism · Skeletal Muscle Relaxants

Which agent is a Schedule IV controlled substance because it is metabolized to meprobamate?

- **A** Chlorzoxazone
- **B** Methocarbamol
- **C** Tizanidine
- **D** Carisoprodol
- **E** Metaxalone

ANSWER: D

Carisoprodol is metabolized to meprobamate and is a Schedule IV controlled substance with dependence/misuse potential (S33). S33

- **A.** Noted for rare hepatotoxicity (S33).
- **B.** Generally less sedating than cyclobenzaprine; not the Schedule IV drug taught (S33).
- **C.** Central α2 agonist; not scheduled in the lecture (S29).
- **E.** Relatively favorable sedation profile (S33).

Q11 · Clinical vignette · Skeletal Muscle Relaxants

A young man with extensive burns needs emergency intubation. Why is succinylcholine avoided?

- **A** It lasts many hours in burned patients
- **B** Potassium release can cause cardiac arrest
- **C** It cannot reach the burned muscle tissue
- **D** Burns make it a nondepolarizing blocker
- **E** It is reversed too quickly by sugammadex

ANSWER: B

Burns and denervation proliferate extrajunctional receptors; succinylcholine then releases potassium, which can cause cardiac arrest (p. 508). p. 508

- **A.** Its block lasts under 10 minutes (p. 507).
- **C.** The danger is hyperkalemia (p. 508).
- **D.** It remains depolarizing (p. 503).
- **E.** Sugammadex reverses rocuronium and vecuronium, not succinylcholine (p. 509).

Q12 · Clinical vignette · Bone Pharmacology & Osteoporosis

A 66-year-old woman starts weekly alendronate. Which counseling point most reduces her risk of esophageal injury?

- **A** Take it with breakfast to protect the stomach
- **B** Take it at bedtime with a calcium supplement
- **C** Lie down for 30 minutes after the dose
- **D** Take it with coffee to speed gastric emptying
- **E** Full glass of plain water, then stay upright 30 minutes

ANSWER: E

Take with a full glass of water first thing in the morning, fasting, and remain upright — this reduces esophageal contact time and irritation (S52; S53; SG U2-R13). S52; S53; SG U2-R13

- **A.** Food cuts the already poor absorption; take it fasting (S53).
- **B.** Calcium blocks absorption, and lying down raises esophageal risk (S53).
- **C.** Remaining upright is the key counseling point (S52).
- **D.** Coffee reduces absorption (S53).

Q13 · Mechanism · Bone Pharmacology & Osteoporosis

Which option correctly describes the PTH feedback loop?

- **A** Low calcium → vitamin D → PTH suppression and calcium excretion
- **B** High calcium → PTH → bone formation, renal excretion, less gut absorption
- **C** Low calcium → PTH → bone resorption, renal reabsorption, vitamin D gut absorption
- **D** Low calcium → calcitonin → bone resorption and renal reabsorption
- **E** Low calcium → PTH → osteoblast-only bone formation

ANSWER: C

Falling calcium triggers PTH, which raises calcium via bone resorption, renal reabsorption, and (via vitamin D activation) intestinal absorption (S43; SG U2-R12). S43; SG U2-R12

- **A.** Not the taught loop (S43).
- **B.** PTH is triggered by low calcium and raises it (S43).
- **D.** Calcitonin lowers calcium and opposes PTH (S42).
- **E.** Physiologic PTH raises calcium partly through osteoclast resorption (S43).

Q14 · Mechanism · Bone Pharmacology & Osteoporosis

Which osteoporosis drug is an anabolic PTH analog that stimulates osteoblast bone formation?

- **A** Teriparatide
- **B** Denosumab
- **C** Alendronate
- **D** Raloxifene
- **E** Calcitonin

ANSWER: A

Teriparatide (PTH 1-34, daily injection) is an anabolic agent that stimulates osteoblasts and new bone formation (S54; S55; SG U2-M). S54; S55; SG U2-M

- **B.** Denosumab is a RANKL inhibitor that blocks osteoclast formation (S54).
- **C.** Alendronate is an antiresorptive bisphosphonate (S47).
- **D.** Raloxifene is an antiresorptive SERM (S55).
- **E.** Calcitonin is antiresorptive and opposes PTH (S55).

Q15 · Clinical vignette · Bone Pharmacology & Osteoporosis

A patient on denosumab for 4 years wants to stop it because of the injections. What should the PA advise?

- **A** Stop and start calcium only
- **B** Take a drug holiday, as for bisphosphonates
- **C** Switch to teriparatide for life
- **D** Don’t just stop — transition to a bisphosphonate
- **E** Stop safely: its effect lasts years in bone

ANSWER: D

Denosumab discontinuation causes rebound vertebral fractures; transition to a bisphosphonate (S55). S55

- **A.** Calcium alone won’t prevent the rebound (S55).
- **B.** Drug holidays apply to bisphosphonates, not denosumab (S48; S55).
- **C.** Teriparatide is limited to 2 years lifetime (S55).
- **E.** That long skeletal half-life describes bisphosphonates (S48).

Q16 · Clinical vignette · Bone Pharmacology & Osteoporosis

A 70-year-old with a myocardial infarction 6 months ago has severe osteoporosis. Which agent should be avoided because of its boxed warning?

- **A** Romosozumab
- **B** Teriparatide
- **C** Alendronate
- **D** Zoledronic acid
- **E** Denosumab

ANSWER: A

Romosozumab carries a boxed warning for MI, stroke, and cardiovascular death (S55). S55

- **B.** Its limit is 2 years lifetime (S55).
- **C.** Bisphosphonate concerns are esophageal, hypocalcemia, ONJ, atypical fracture (S52).
- **D.** Its issues are the acute-phase reaction and renal limits (S48; S53).
- **E.** Its key warning is rebound fractures on stopping (S55).

Q17 · Clinical vignette · Bone Pharmacology & Osteoporosis

A patient with CrCl 25 mL/min has osteoporosis. Which drug class is contraindicated at this kidney function?

- **A** Sclerostin antibodies such as romosozumab
- **B** RANKL antibodies such as denosumab
- **C** Bisphosphonates such as alendronate
- **D** PTH analogs such as teriparatide
- **E** Estrogen modulators such as raloxifene

ANSWER: C

Bisphosphonates are contraindicated with severe renal impairment (CrCl below 30–35 mL/min) (S48); renal function is part of choosing therapy (S59). S48; S59

- **A.** Romosozumab's concern is the CV boxed warning (S55).
- **B.** The renal cutoff taught is for bisphosphonates (S48).
- **D.** Their limit is 2 years of use (S55).
- **E.** Raloxifene's concern is VTE (S55).

Q18 · Mechanism · Bone Pharmacology & Osteoporosis

Why does raloxifene suit some patients with vertebral osteoporosis but not those whose main risk is hip fracture?

- **A** It is contraindicated in all women over 65
- **B** It must be stopped after 2 years
- **C** It builds new bone only in the hip
- **D** It raises hip fracture risk through VTE
- **E** It reduces spine fractures but not hip fractures

ANSWER: E

Raloxifene protects against spine fractures but not hip fractures, unlike bisphosphonates, denosumab, and teriparatide (p. 827). p. 827

- **A.** Not a taught contraindication (p. 827).
- **B.** The 2-year limit is teriparatide (p. 827).
- **C.** It is antiresorptive and spine-protective (p. 827).
- **D.** VTE is a clotting risk, not a fracture mechanism (p. 827).

Q19 · Clinical vignette · Putting It All Together: Musculoskeletal Toolkit

A 72-year-old with OA, CKD, hypertension, and an anticoagulant now has an acute gout flare. Which plan best fits the study guide?

- **A** Full-dose colchicine with no adjustment
- **B** Start febuxostat today to stop the flare
- **C** Probenecid now, since the kidneys are the issue
- **D** Full-dose oral naproxen; start allopurinol now
- **E** Short corticosteroid course; defer allopurinol

ANSWER: E

An oral NSAID is relatively contraindicated (CKD, anticoagulant, hypertension); a short corticosteroid course is often preferred, colchicine needs renal dose adjustment, and ULT waits until the flare resolves (SG U2-S). SG U2-S; S9; S11

- **A.** Colchicine needs renal dose adjustment (S11).
- **B.** ULT never treats a flare (S9).
- **C.** Probenecid is chronic ULT and avoided in renal impairment (S14).
- **D.** NSAID red flags stack up, and ULT waits (SG U2-S).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO2.1** Identify major pharmacologic classes used in the management of musculoskeletal conditions and describe their mechanisms of action.
- [ ] **IO2.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat musculoskeletal conditions.
- [ ] **IO2.3** Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.
- [ ] **IO2.4** Select appropriate pharmacologic therapy for common musculoskeletal conditions based on clinical presentation and patient-specific factors.
- [ ] **IO2.5** Identify appropriate dosing, administration, and monitoring considerations for medications used in musculoskeletal conditions.
- [ ] **IO2.6** Explain important medication-use and safety information to patients receiving pharmacotherapy for musculoskeletal.

- [ ] **SO1** Explain the pharmacologic treatment of acute and chronic gout, including representative generic agents.
- [ ] **SO2** Classify skeletal muscle relaxants by mechanism and clinical use, including representative generic agents.
- [ ] **SO3** Explain the pharmacologic management of osteoporosis, including representative generic agents.
- [ ] **SO4** Identify key contraindications, adverse effects, and drug interactions throughout.
- [ ] **SO5** Apply the full musculoskeletal pharmacology toolkit to integrated clinical scenarios.